

Geometric algebra as the input language of collider foundation models

E. Abasov¹, L. Dudko^{1*}, F. Grigoryev¹, P. Volkov¹ and A. Zaborenko¹

¹ Skobeltsyn Institute of Nuclear Physics, Lomonosov Moscow State University (SINP MSU),
1(2) Leninskie gory, GSP-1, Moscow 119991, Russian Federation

★ dudko@sinp.msu.ru

Abstract

A hard hadron-collider event is represented here as a single geometric object: the kinematics and object-type labels of all reconstructed final-state particles in one multivector $\mathcal{E}$, one $\text{Cl}(1,3) \otimes V_{\text{flav}}$ slot per object, whose higher grades the algebra generates from the measured four-momenta — rather than as a list of four-momenta with label fields attached. The setting is geometric algebra [1–3], whose grade decomposition organises essentially every observable in current use: invariants at grade zero, four-momenta at grade one, decay-plane bivectors at grade two, oriented three-volumes at grade three, the CP-odd pseudoscalar at grade four. The high-level invariants of [4], the low-level recipe of [5] and the equivariant-network inputs of [6–10] are recovered as projections onto specific grades. A per-grade dictionary of 32 classical observables and an inventory of the symmetries acting on $\mathcal{E}$ are provided. Theorem 1 settles which Lorentz invariants the higher grades unlock: none beyond $\{p_i \cdot p_j, m_i^2\}$, the CP-odd sign of the pseudoscalar being the one genuine channel. The representation is intended as a uniform input layer for foundation models of collider physics, and the realisations demonstrated here occupy two rungs of a ladder: the higher grades are formed inside the network from the per-object slots, or selected grade-two and grade-three objects are supplied at the input layer. It is demonstrated on the resonance-topology separation of $pp \rightarrow t W b$ [11–13] with a Lorentz-equivariant multivector transformer of L-GATr type [10]. An ablation over eight input representations separates what supplying algebraic content explicitly can and cannot buy: content reconstructible from the measured momenta unlocks no new invariant, by the theorem, whereas a fixed reference blade encoding the beam plane carries structure the momenta do not contain. Which sectors a task reads is a property of the process, so the representation carries them all.

Copyright attribution to authors.

This work is a submission to SciPost Physics.

License information to appear upon publication.

Publication information to appear upon publication.

Received Date

Accepted Date

Published Date

Contents

1	Introduction	2
2	Geometric algebra in $\text{Cl}(1,3)$	4
2.1	Relation to symmetry-aware architectures	5
2.2	The event multivector	6

3	Physical meaning of multivector grades	7
4	Operations on the event multivector	12
5	Symmetries and equivariance	15
6	Object-type space V_{flav}	17
7	Resonance-topology separation in $pp \rightarrow tWb$	20
7.1	Which grades earn their place: an input ablation	25
8	Foundation models for collider events	29
8.1	Practical obstacles	31
9	Limitations and future directions	32
10	Summary	34
A	Notation card	34
B	Algebraic derivations	36
C	Tables	39
C.1	Full per-grade dictionary of classical observables	39
C.2	Symmetry inventory	41
C.2.1	Continuous spacetime symmetries	42
C.2.2	Discrete spacetime symmetries	42
C.2.3	Internal and permutational symmetries	42
C.2.4	Approximate, ML-architectural and CP-channel symmetries	43
D	Token layouts of the two input branches	44
	References	45

1 Introduction

The performance and interpretability of a neural network trained on collider data depend on what it is allowed to *see* — on the input representation — at least as much as on the architecture itself.

The event as a single geometric object. The first message of this paper is that a hard hadron-collider event should be regarded as a single geometric object — one graded multivector per event — rather than as a list of four-momenta accompanied by discrete labels for charge, b -tag, lepton flavour and missing transverse momentum. This is a statement about the representation, realised operationally in Sec. 2.2; which grades a given network is handed at its input layer, rather than building them itself, is an implementation choice measured in Sec. 7.1. The existing high-level invariant recipes [4], low-level four-momentum recipes [5], and the equivariant networks of refs. [6–10, 14] all act on disjoint subsets of this object: the first two answer *which scalar features to feed the network*, the third answers *how to make the*

network respect Lorentz symmetry without dictating the features, and in all three the discrete reconstructed information sits outside the kinematic vector space.

Geometric algebra is the natural framework. The second message is that geometric algebra [1–3] is the natural mathematical framework for this view, and that essentially all observables routinely used in collider analyses are already organised by its grade decomposition. A reconstructed event is represented as a single multivector $\mathcal{E}$, defined in Eq. (4) with one $\text{Cl}(1, 3) \otimes V_{\text{flav}}$ slot per reconstructed object. The Lorentz-invariant scalar inputs of refs. [4, 5] are recovered as the grade-zero sector of $\mathcal{E}$; lab-frame and helicity-frame angular variables sit in the beam-tensored grade-one sector together with the explicit reference vectors that define each frame; the raw covariant coefficients of the higher-grade components — already adopted as the input channels of L-GATr [10] — are the natural input layer for Lorentz-equivariant networks. The construction is a common notational framework, not new physical content.

Towards foundation models for collider events. The third message is that this representation is the substrate on which a foundation model for collider physics can be built: a single multivector type $\mathcal{E}$, the same per-grade decomposition across processes, and a uniform fine-tuning interface to classification, regression and CP-asymmetry tasks (Sec. 8).

From conceptual framework to practical architecture. The conceptual content of this paper — the unified algebraic representation, the result that the higher grades unlock no new Lorentz-invariant scalar (Theorem 1, which follows because the squared magnitude of a k -blade is a Gram determinant of inner products) and the grade-resolved fine-tuning interface — is independent of any particular network. Throughout, a *token* is one input slot of the network, holding a stack of channels; each channel is a full 16-component multivector of $\text{Cl}(1, 3)$, whose $1+4+6+4+1$ grade cells are *storage* rather than content. The particle tokens are the slots of eq. (4) themselves; event-level and reference tokens extend the sequence in the same format. Which cells arrive populated is precisely the input choice under test (App. D gives the layouts number by number). The architectural content sits in Sec. 7, where the $pp \rightarrow tWb$ demonstration combines two ingredients. The first is per-particle multivector tokens — one multivector T_i per reconstructed object i , in the manner of L-GATr [10] — encoding each object as $T_i = \langle T_i \rangle_0 \oplus \langle T_i \rangle_1$, writing $\langle X \rangle_k$ for the grade- k part of a multivector X . The second ingredient is event-level pairing tokens that append grade-two and grade-three candidate-pairing content of the multi-resonance topology at the input layer, realising the event-as-multivector view at the input rather than only through internal self-attention. The ablation of Sec. 7.1 measures what each of these input choices is worth on this benchmark, and the outcome is not the one we anticipated: the topology tokens do not help, while a fixed grade-two reference bivector encoding the beam plane does. The benchmark was chosen deliberately minimal — the entire class difference is a single resonant three-particle combination, one grade-three object — so pruning everything else can only help this particular task, and the measurement must not be read as a verdict on explicit higher-grade inputs in general. What it does establish is a distinction between algebraic content derivable from the measured momenta and content that is not; which derived components matter is a property of the process, and a representation intended to serve many processes must carry the full graded inventory precisely because each task reads a different sector of it.

Why geometric algebra. Three properties of $\text{Cl}(1, 3)$ explain why a single algebraic framework, rather than a feature concatenation across recipes, is the right input for a foundation model of collider events. (i) *Closure under $\text{Spin}^+(1, 3)$ — the group of even, unit-norm elements of the algebra, the double cover of the proper orthochronous Lorentz group, acting on multivectors*

by the two-sided rotor product — at the input level: every equivariant bilinear map on multivector inputs is a grade projection of the geometric product (Sec. 4), so the algebra carries the inductive bias of Lorentz-covariant networks without auxiliary machinery. (ii) *No new invariants above grade one*: the higher-grade magnitudes add no Lorentz-invariant scalar beyond the inner products and masses $\{p_i \cdot p_j, m_i^2\}$ (Theorem 1); the only genuinely new Lorentz-invariant content is the one-bit CP-odd sign of the pseudoscalar. (iii) *Per-grade pre-training surface*: the grade decomposition (eq. (3)) supplies a fixed grade interface for every downstream task (Sec. 8).

The collapse theorem underlying point (ii) extends the classical Euclidean Cayley–Menger / Gram-determinant identity [15, 16] to the Lorentzian signature, in the spirit of the simplex-realizability analysis of ref. [17] (Theorem 1); the relation of the resulting CP-odd sign to earlier CP analyses [18] is discussed in Sec. 3.

The remaining content of the paper is organised around $\mathcal{E}$ as the input specification of a foundation-model program for collider events [19]: an explicit per-grade dictionary of classical observables, the discrete-symmetry catalogue of the object-type factor V_{flav} , and a grade-resolved pre-training interface to downstream tasks (Sec. 8).

Sec. 2 fixes conventions; Sec. 3 works through the grade-by-grade dictionary of the kinematic factor; Sec. 4 catalogues the operations on $\mathcal{E}$ with their physical interpretation; Sec. 5 treats the symmetry inventory; Sec. 6 catalogues the discrete object-type factor V_{flav} of the event multivector and the handling of missing transverse momentum; Sec. 7 presents the $pp \rightarrow tWb$ demonstration; Sec. 8 outlines the foundation-model program; Sec. 9 discusses limitations and future directions; Sec. 10 summarises the contributions.

2 Geometric algebra in $\text{Cl}(1, 3)$

The remainder of the paper builds on a small subset of the geometric algebra $\text{Cl}(1, 3)$ of Minkowski spacetime [1–3]. We restate here exactly the facts that we use later, in the form most directly tied to the multivector representation of a collider event. A reader familiar with $\text{Cl}(1, 3)$ may skim this section and consult Appendix A for the full notation card; algebraic identities used in the body are derived in Appendix B.

Convention block. The Clifford algebra $\text{Cl}(1, 3)$ of Minkowski spacetime is used throughout, with metric signature $(+, -, -, -)$ and an orthonormal grade-one basis $\{\gamma_0, \gamma_1, \gamma_2, \gamma_3\}$ satisfying

$$\gamma_\mu \gamma_\nu + \gamma_\nu \gamma_\mu = 2\eta_{\mu\nu}, \quad \eta = \text{diag}(+, -, -, -). \quad (1)$$

The geometric product of two grade-one elements splits into a symmetric inner part reproducing the Minkowski metric, $pq = \frac{1}{2}(pq + qp)$, and an antisymmetric outer part, $p \wedge q = \frac{1}{2}(pq - qp)$. The explicit decomposition $pq = p \cdot q + p \wedge q$ is eq. (A.1), with component form eq. (A.2). The notation card collects, once and for all, the basis conventions, the Hodge dual $\star$, the bivector sign rules and the grade-projection notation used throughout; The proper orthochronous Lorentz group acts on multivectors through the rotor sandwich

$$X \mapsto RX\tilde{R}, \quad R = \exp(B/2), \quad B \in \langle \text{Cl}(1, 3) \rangle_2, \quad R\tilde{R} = 1, \quad (2)$$

with R in the spin cover $\text{Spin}^+(1, 3) \simeq \text{SL}(2, \mathbb{C})$. The pseudoscalar $I = \gamma_0\gamma_1\gamma_2\gamma_3$ ($I^2 = -1$), the reverse $\tilde{X}$, the grade projection $\langle X \rangle_k$, the Hodge-type dual $X^\star \equiv X I^{-1}$ and the corresponding inverse-dual $X^{-\star} \equiv X I$, together with the bivector sign rules and the canonical decomposition of a generic bivector into commuting rotation and boost planes, are collected in the notation card.

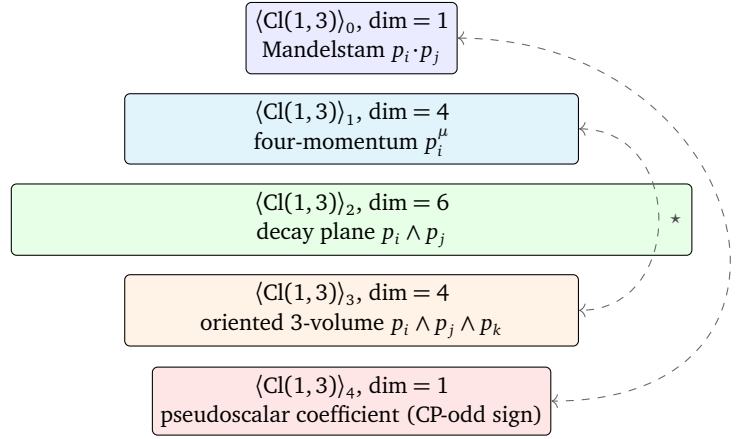


Figure 1: Grade ladder of $\text{Cl}(1, 3)$. The five homogeneous components have dimensions $(1, 4, 6, 4, 1)$, the binomial coefficients summing to $2^4 = 16$; each grade carries a definite physical reading on collider four-momenta. The Hodge dual $X^\star \equiv X I^{-1}$ pairs grade k with grade $4 - k$ (dashed arcs on the right).

The algebra $\text{Cl}(1, 3)$. The algebra splits into homogeneous components by grade,

$$\text{Cl}(1, 3) = \bigoplus_{k=0}^4 \langle \text{Cl}(1, 3) \rangle_k, \quad \dim \langle \text{Cl}(1, 3) \rangle_k = \binom{4}{k}, \quad 1 + 4 + 6 + 4 + 1 = 16. \quad (3)$$

Each grade carries a direct physical reading (Fig. 1): grade zero is a Lorentz scalar, e.g. a Mandelstam invariant $p_i \cdot p_j$; grade one is a four-momentum; grade two is an oriented decay plane $p_i \wedge p_j$; grade three is an oriented 3-volume $p_i \wedge p_j \wedge p_k$, equivalently a covariant four-vector via the dual $\star$; grade four is the pseudoscalar coefficient $\epsilon_{\mu\nu\rho\sigma} p_1^\mu p_2^\nu p_3^\rho p_4^\sigma$, with $\epsilon_{\mu\nu\rho\sigma}$ the totally antisymmetric Levi-Civita symbol; it is a CP-odd one-bit observable through its sign, since an odd number of index permutations — equivalently a spatial inversion — reverses it. The squared magnitude of any r -blade equals the $r \times r$ Gram determinant of the inner products of the participating four-momenta (Theorem 1); the meet $\cap$ and the join $\cup$, derived from the dual and the outer product, are treated together with the other operations on $\mathcal{E}$ in Sec. 4. We adopt the high-energy convention $(+, -, -, -)$ throughout; the alternative signature $(-, +, +, +)$ realises the algebra $\text{Cl}(3, 1) \simeq M_4(\mathbb{R})$, which is physically equivalent to $\text{Cl}(1, 3) \simeq M_2(\mathbb{H})$ at the level of observable quantities — the two algebras share the same even subalgebra $\text{Cl}^+(1, 3) \simeq \text{Cl}^+(3, 1)$ and the same spin cover $\text{Spin}^+(1, 3) \simeq \text{SL}(2, \mathbb{C})$ — but they are not isomorphic as real associative algebras [3, Tab. 16.4], and we keep $\text{Cl}(1, 3)$ for consistency with the high-energy literature. The conformal extension $\text{Cl}(2, 4)$ is a distinct algebra that lifts the grade bound at the cost of two extra basis vectors, deferred as an algebraic extension in Sec. 9.

2.1 Relation to symmetry-aware architectures

Where a network places the symmetry is a design choice with three established answers, and the present work is a statement about the first of them, so it is worth separating them before the construction begins.

Invariance in the aggregation. The PELICAN family [7, 20] feeds a network the Lorentz invariants of the input — the array of inner products — and aggregates permutation-equivariantly. LorentzNet [21] builds its messages from the same invariants but carries equivariant four-vector channels through its layers, placing it between this family and the next. The result is

exactly invariant by construction and needs no equivariant layers. Its ceiling is set by Theorem 1: what such a network cannot see is the one channel not reducible to inner products, the orientation sign of the grade-four pseudoscalar. This line has a direct internal analogue in our study, the grade-zero-only rung of Tab. 3, which loses 0.00422 ± 0.00025 in area under the ROC curve to grade-0 $\oplus$ 1 four-momenta in the same architecture (151 489 against 149 409 parameters — the invariants-only network is the marginally larger one, so capacity cannot explain the loss) — a measurement of what the covariant input buys over invariants alone, with everything else held fixed.

Equivariance in the layers. The Lorentz Group Network (LGN) [6], Clifford-group equivariant networks [22], $E(n)$ -equivariant graph networks [23] and the Lorentz-equivariant geometric-algebra transformer line [8–10] build the group action into the weights, so covariant quantities may be carried through the network. These are the closest neighbours of the present work and we differ from them in emphasis, not in mechanism: they specify layers, we specify what the input tensor should contain, and Sec. 7.1 is a measurement of that specification with the layers held fixed. L-GATr [10] already carries fixed reference multivectors alongside the per-particle ones, which is precisely the ingredient our ablation finds to be the useful one.

Symmetry left to the data. Deep-sets architectures on infrared-safe observables [24], point-cloud convolutions with augmentation [25], and attention with a pairwise-invariant bias [26, 27] impose no exact group action and are competitive on jet-level benchmarks. They are not alternatives to a representation claim but a reminder of its scope. An inductive bias has to earn its place against a model that simply learns the symmetry, which is why Sec. 7.1 reports the outcome for our own higher-grade inputs even though it is unfavourable.

A fourth line addresses the combinatorial problem our demonstration exposes rather than the symmetry: symmetry-preserving attention networks assign jets to parents directly [28, 29], without any algebraic structure. We do not benchmark against them. The reason is not cost — the ablation of Sec. 7.1 is forty trainings of this network. It is that the two tasks do not share a figure of merit: those networks are scored on assignment accuracy against a truth permutation, whereas the demonstration here is a topology classification in which no assignment label appears. Retargeting their architecture to our objective would measure our tuning of someone else’s model. The comparison that can be made cleanly is the one we make, varying the input representation with architecture, objective and data fixed.

2.2 The event multivector

For a reconstructed event with N final-state detector objects indexed by $i = 1, \dots, N$, with four-momenta $p_i \in \text{Cl}(1, 3)$ and discrete object-type labels $f_i \in \mathbb{F}$, the *event multivector* is the direct sum

$$\mathcal{E} \equiv \bigoplus_{i=1}^N p_i \otimes |f_i\rangle \in [\text{Cl}(1, 3) \otimes V_{\text{flav}}]^{\oplus N}, \quad (4)$$

where $V_{\text{flav}} = \text{span}_{\mathbb{R}}\{|f\rangle : f \in \mathbb{F}\}$ is a finite-dimensional real vector space, trivial under the Lorentz group, that records the discrete object-type label of each detector object. The direct sum runs over object slots i , one copy of $\text{Cl}(1, 3) \otimes V_{\text{flav}}$ per reconstructed object, so two objects sharing a type label keep their individual momenta. Implementations realise the slots as the N input tokens of the network.

Equation (4) is the grade-one ingredient only: a typed list of four-vectors, and as a direct sum it carries no product between different slots. What supplies one is that every slot’s kinematic factor lives in one and the same copy of $\text{Cl}(1, 3)$: writing $\pi_i : \mathcal{E} \mapsto p_i$ for the projection onto slot i , products across slots are formed in that common algebra, and a monomial in k distinct slots is labelled by the corresponding element of $V_{\text{flav}}^{\otimes k}$. The *single geometric object* of

the introduction is accordingly the subspace

$$\bar{\mathcal{E}} = \bigoplus_{k \geq 1} \text{span} \{ \pi_{i_1} \mathcal{E} \cdots \pi_{i_k} \mathcal{E} \otimes |f_{i_1}\rangle \otimes \cdots \otimes |f_{i_k}\rangle : i_1 < \cdots < i_k \},$$

the closure of the slots under the Clifford operations of Sec. 4 applied on the $\text{Cl}(1, 3)$ factor, with the V_{flav} tags carried along as passive labels. It is a subspace of $\text{Cl}(1, 3) \otimes V_{\text{flav}}^{\otimes \bullet}$, indexed by subsets of the slots. Wedging the p_i pairwise generates the $\binom{N}{2}$ decay-plane bivectors at grade two (the leptonic- W plane $p_\ell \wedge p_\nu$ of the tWb example below is one such blade), triple wedges yield the $\binom{N}{3}$ oriented three-volumes at grade three, and quadruple wedges yield the $\binom{N}{4}$ pseudoscalar coefficients — CP-odd one-bit observables — at grade four; inner products of the same p_i supply the grade-zero Lorentz scalars (resonance masses, Mandelstam variables, helicity-frame projections), and bivector exponentials $\exp(B/2)$ generate the rotors of Sec. 5 that implement boosts and rotations on the object. The combinatorial filling of all five grades from the single grade-one input is drawn for $N = 6$ in Fig. 2; the resulting $\text{Cl}(1, 3) \otimes V_{\text{flav}}$ factorisation that the network sees is summarised in Fig. 3. The event is a single geometric object in this operational sense: not a list of four-momenta with side labels, but the graded multivector formed by the algebraic closure of the typed grade-one input under the operations of Sec. 4. Which grades a network receives *explicitly* is a separate, implementation-level choice, and the realisations form a ladder. At one end the network sees only the grade-one summands and builds the closure internally through its equivariant layers; this is the L-GATr-type configuration of Sec. 7. Theorem 1 guarantees that no Lorentz-invariant information is lost on this rung, and the ablation of Sec. 7.1 finds that on its deliberately minimal benchmark explicit materialisation of the derived components does not pay. Which components a task needs is process-dependent, so the representation itself carries all five grades; how many of them to materialise explicitly is a per-task, per-architecture choice that the ladder is built to measure. At the other end the whole inventory is materialised as its own tokens before the network sees it, which is the form the pre-training interface of Sec. 8 assumes and which no implementation, ours included, yet supplies in full. The distinction between the rungs is *where* the higher grades are formed, not whether the event is one object: they are joint content of several particles at every rung — $p \wedge p = 0$, so a single particle contributes nothing above grade one — and the graded object they populate is the event throughout. The present paper is the specification of that object together with a measurement of the first rungs of the ladder. Throughout the paper $\langle \mathcal{E} \rangle_k$ denotes the grade- k sector of this algebraic closure (the V_{flav} tags carried along as passive labels); eq. (4) itself populates only grade one.

3 Physical meaning of multivector grades

In this section we develop the physical meaning of each grade of the kinematic factor $\text{Cl}(1, 3)$ of the event multivector $\mathcal{E}$ defined in Sec. 2.2 (eq. (4)); the discrete object-type factor V_{flav} (eq. (14)) and the missing-transverse-momentum pseudo-object (eq. (15)) are catalogued in Sec. 6. The grade decomposition of the Clifford algebra $\text{Cl}(1, 3) = \bigoplus_{k=0}^4 \langle \text{Cl}(1, 3) \rangle_k$ admits a direct physical reading on the four-momenta of an event, and that reading carries through to the multivector $\mathcal{E}$ of eq. (4) slot by slot. The five subsections that follow give the algebraic object, its physical interpretation, the classical observables that it correlates with through the prior input recipes of refs. [4, 5], and the one-line statement of what is genuinely unique to the multivector representation versus what already follows from the grade-zero sector through Theorem 1. Tab. 1 summarises the per-variable mapping in 18 rows; the full 32-row dictionary, with the operation column expanded to a Cayley–Menger status, is collected in App. C.1. Fig. 2 shows how, for an $N = 6$ semi-leptonic $t\bar{t}$ -like final state, the per-particle four-momenta

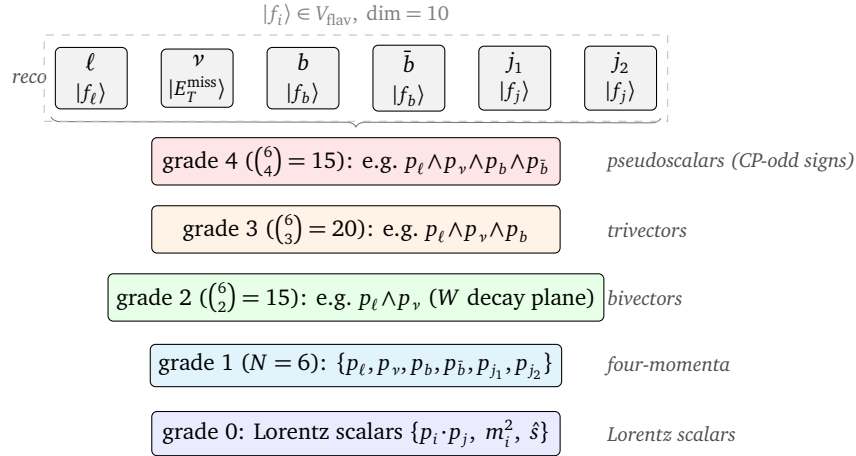


Figure 2: Per-particle assembly of the event multivector for an $N = 6$ reconstructed final state $\{\ell, \nu, b, \bar{b}, j_1, j_2\}$ typical of semi-leptonic $t\bar{t}$ (charged lepton, neutrino, two b -tagged jets, two light jets). Each object carries a four-momentum p_i (grade 1 in $\text{Cl}(1,3)$) tensored with a flavour tag $|f_i\rangle \in V_{\text{flav}}$ from the discrete object-type space; the basis of V_{flav} enumerates the ten reconstructed object types of eq. (14), so $\dim V_{\text{flav}} = 10$ is set by the object vocabulary and is independent of the per-event count N . The five grades of $\mathcal{E}$ are populated combinatorially: $N = 6$ grade-one four-vectors, $\binom{6}{2} = 15$ grade-two bivectors (decay planes such as $p_\ell \wedge p_\nu$ for the leptonic W), $\binom{6}{3} = 20$ grade-three trivectors, $\binom{6}{4} = 15$ grade-four pseudoscalars, and the grade-zero sector containing all Lorentz scalars built from inner products $\{p_i \cdot p_j, m_i^2, \hat{s}\}$. Every object enters every grade, grade k collecting all $\binom{N}{k}$ wedge products, which is why the brace groups the objects rather than lines being drawn object by object. One explicit example is shown per higher grade; counts on the right. The parton-level neutrino is carried in the reconstruction vocabulary as the E_T^{miss} pseudo-object of eq. (15).

combinatorially populate every grade of $\mathcal{E}$ through their wedge products.

Grade 0 — Lorentz scalars. The grade-zero sector $\langle \mathcal{E} \rangle_0$ collects the Lorentz-invariant numbers built from the four-momenta of the event. All of them are polynomials in the inner products $p_i \cdot p_j$: the on-shell masses $m_i^2 = p_i \cdot p_i$ on the diagonal, and the invariant mass of any subset, $m_S^2 = (\sum_{i \in S} p_i)^2$, off it. The Mandelstam variables are the same construction applied to particular subsets — $\hat{s}$ to the full final state, $\hat{t}$ and $\hat{u}$ to the crossed combinations with the initial partons, which are carried as two further slots of eq. (4) with their own V_{flav} labels whenever the crossed invariants are used — so the resonance masses m_{ij} , m_{ijk} and the partonic $\hat{s}$ are one family, not several.

Physically the grade-zero sector is the resonance ladder of the event: peaks of m_{ij} at W, Z, H masses, peaks of m_{ijk} at t and $\bar{t}$, and the partonic $\sqrt{\hat{s}}$ that controls threshold behaviour and parton-luminosity weighting. The spin-analysing angular projections $\cos \theta^*$ — the lepton angle in the top rest frame and its optimal-basis variants, introduced for single-top and $t\bar{t}$ spin correlations by Mahlon and Parke [30–32] and used in the input recipe of ref. [4] — together with the Collins–Soper angle $\cos \theta_{CS}$, are Lorentz scalars constructed by a rotor sandwich $X \mapsto R X \tilde{R}$ followed by an inner product, and therefore live in $\langle \mathcal{E} \rangle_0$ as well.

By construction $\langle \mathcal{E} \rangle_0$ contains all Lorentz-invariant scalar inputs of the prior input recipes of refs. [4, 5]: the recipe of ref. [5] keeps the pairwise inner products and $\hat{s}$ explicitly; the high-level reconstructed masses of ref. [4] are linear combinations of the same set. Cf. Tab. 1

rows 1–4 for the per-variable mapping.

Theorem 1 (Cayley–Menger collapse, Lorentzian extension). *Let $p_1, \dots, p_N \in \text{Cl}(1, 3)$ be grade-one elements and let $X_1, \dots, X_n$ be homogeneous-grade multivectors, each an outer product of the $\{p_i\}$. Write λ_{ijkl} for the pseudoscalar coefficient of $\langle p_i p_j p_k p_l \rangle_4 = \lambda_{ijkl} I$ (eq. (8)). Then every Lorentz-invariant scalar $\langle X_1 X_2 \dots X_n \rangle_0$ is a polynomial in the inner products $\{p_i \cdot p_j\}_{i,j=1}^N$ and the coefficients λ_{ijkl} , of degree at most one in the λ 's jointly: it has the form $P + \sum_{i < j < k < l} \lambda_{ijkl} Q^{ijkl}$ with P and Q^{ijkl} polynomials in the inner products alone, since every product $\lambda_{ijkl} \lambda_{i'j'k'l'}$ is itself a polynomial in the inner products. In particular the reversion contraction of any r -blade, $r = 2, 3, 4$, equals the $r \times r$ Gram determinant, $\langle P_r \tilde{P}_r \rangle_0 = \det(p_{i_a} \cdot p_{i_b})$ (eqs. (B.1), (7), (B.3)).*

Corollary. The Gram matrix determines every Lorentz-invariant scalar up to one bit. Under a spatial inversion all λ_{ijkl} change sign together while the inner products are unchanged, and any ratio $\lambda_T / \lambda_{T'}$ is fixed by the Gram matrix through $\lambda_T \lambda_{T'} = -\det(p_a \cdot p_b)_{a \in T, b \in T'}$; the single common orientation sign is therefore the only invariant content not reducible to $\{p_i \cdot p_j\}$.

The proof is given in App. B. The polynomial part is the First Fundamental Theorem of invariant theory for the orthogonal group [33, Ch. II] specialised to $\text{SO}(1, 3)$; the Gram-determinant identity is the Lorentzian-signature form of the Euclidean Cayley–Menger identity [15, 16], used for $(n, 1)$ -simplex realizability in [17, App. A]. The Clifford-algebraic formulation adds one thing: it exhibits the unique non-polynomial invariant — the orientation bit — as a single channel of the event multivector $\mathcal{E}$.

The consequence for the input layer is what Sec. 7.1 measures: a grade- k object built by wedging the measured momenta contributes no Lorentz-invariant information beyond their Gram matrix, however explicitly it is presented to a network.

The non-trivial Lorentz-invariant content of the higher grades is therefore the pseudoscalar sign alone, discussed in the grade-four paragraph below.

Grade 1 — four-vectors. The grade-one sector $\langle \mathcal{E} \rangle_1$ is spanned by the per-particle four-momenta p_i and by their sums (see Fig. 2, second row from the bottom), in particular the total event four-momentum $p_{\text{tot}} = \sum_i p_i$. The transverse momentum p_T^f , the rapidity y^f and the azimuth ϕ^f of object f are not Lorentz scalars: they require a fixed beam-axis reference vector $\hat{n}_z$ and a fixed transverse plane. We make this dependence visible by writing them as components of the grade-one element p_f in the beam-axis-adapted representation

$$p_f = (E^f, p_x^f, p_y^f, p_z^f) = (M_T^f \cosh y^f, p_T^f \cos \phi^f, p_T^f \sin \phi^f, M_T^f \sinh y^f), \quad (5)$$

with $M_T^f \equiv \sqrt{m_f^2 + (p_T^f)^2}$. The two parameterisations $(E, \vec{p})$ and (p_T, η, ϕ, M) are related by eq. (5) and refer to the same grade-one object in two different bases. In the massless limit the rapidity y reduces to the pseudorapidity $\eta = -\ln \tan(\theta/2)$, which is the collider-standard input variable when the per-object mass is small or unmeasured.

Physically the grade-one sector carries the explicit kinematic content of every detector object. The lab-frame coordinates (p_T, η, ϕ) separate the longitudinal-boost-invariant content (M_T, y) from the azimuthal-rotation-invariant content (p_T, ϕ) , and this separation is precisely the one realised by the algebraic split $\text{Cl}(1, 3) \cong \text{Cl}(1, 1)_z \hat{\otimes} \text{Cl}(0, 2)_\perp$ of Sec. 5, eq. (12), where the graded tensor product $\hat{\otimes}$ is defined as the $\mathbb{Z}_2$ -graded tensor of Clifford algebras. Spin-correlation analyses access grade-one via helicity-frame projections of leptons and b -jets through the rotor sandwich of eq. (2), returning the grade-zero scalar invariants discussed above in the grade-zero paragraph.

The classical event-level constructions reduce to a small list of grade-one outputs. Per-object: $E, p_x, p_y, p_z, p_T, \eta, \phi, y, M$. Event-level: $H_T = \sum_{\text{jets}} p_T, S_T = H_T + \sum_\ell p_T^\ell + E_T^{\text{miss}}$.

Pair-level: $\Delta\eta_{ij}, \Delta\phi_{ij}, \Delta R_{ij} = \sqrt{\Delta\eta_{ij}^2 + \Delta\phi_{ij}^2}$, m_{ij} (the latter through grade-zero, cf. the grade-zero paragraph above). Frame-derived: θ_ℓ^* in the helicity frame through a rotor-sandwich projection. The transverse mass m_T is a grade-one scalar that is invariant only under the LHC residual subgroup H_{LHC} , not under full Lorentz. The missing transverse momentum E_T^{miss} is treated as an incomplete grade-one element with masked longitudinal component and an entry in V_{flav} (cf. eq. (15)). For neural-network input we embed the azimuthal angle as $(\cos\phi, \sin\phi)$ in place of the raw value.¹

Nothing is uniquely new in $\langle\mathcal{E}\rangle_1$: it is the familiar four-vector input of refs. [5, 10]. What is new is its position inside $\mathcal{E}$ as one homogeneous component of a graded multivector rather than as a free-standing object: the geometric product mixes it with the higher grades, and the rotor sandwich preserves the grade decomposition, so a $\text{Spin}^+(1, 3)$ -equivariant network can act on the full grade spectrum without auxiliary combining of channels.

Grade 2 — bivectors and decay planes. The grade-two sector of the kinematic factor, $\langle\text{Cl}(1, 3)\rangle_2$, is six-dimensional. In the orthonormal basis $\{\gamma_\mu\}_{\mu=0,1,2,3}$ the basis splits into three boost-bivectors γ_{0i} ($i = 1, 2, 3$) with $\gamma_{0i}^2 = +1$ and three rotation-bivectors γ_{ij} ($1 \leq i < j \leq 3$) with $\gamma_{ij}^2 = -1$. The simple two-blade $B_{ij} = p_i \wedge p_j$ has square $B_{ij}^2 = (p_i \cdot p_j)^2 - p_i^2 p_j^2$ (eq. (B.1)); for two future-directed time-like momenta the reverse Cauchy–Schwarz inequality $(p_i \cdot p_j)^2 \geq p_i^2 p_j^2$ forces $B_{ij} \widetilde{B_{ij}} \leq 0$, so $|B_{ij}|^2 \equiv -B_{ij} \widetilde{B_{ij}}$ is a non-negative Lorentz scalar.

Physically a unit two-blade $\hat{B} = B/|B|$ is the oriented two-plane in Minkowski spacetime that contains the rest frames of its two participants. After normalisation it is the boost generator that relates these two rest frames at the rapidity $\alpha = \text{arccosh}(p_i \cdot p_j / \sqrt{p_i^2 p_j^2})$ for two on-shell time-like momenta (cf. [1, § 5.107, eq. 5.125]); the limiting cases of light-like or mixed-signature participants are obtained from the same parameterisation by taking the appropriate rapidity limit. The rotor sandwich $B \mapsto R B \bar{R}$ keeps the blade form, so the bivector is a primitive carrier of decay-plane orientation in any equivariant network. The combinations $B_W = p_\ell \wedge p_\nu$ and $B_t = (p_\ell + p_\nu) \wedge p_b$ are the oriented decay planes of the leptonic W and the leptonic t [11].

The classical observables that correlate with grade two are the spin-correlation tensors and the bivector inner products. The Bernreuther basis $\{\hat{k}, \hat{r}, \hat{n}\}$ [34, 35] decomposes the $t\bar{t}$ spin-density matrix into nine coefficients $C_{ij} = -9\langle(\hat{n}_i \cdot \hat{\ell}_+)(\hat{n}_j \cdot \hat{\ell}_-)\rangle$. In the multivector representation these are the grade-zero projections $\langle B_a \widetilde{B_b} \rangle_0$ of the unit two-blades $\hat{B}_t, \hat{B}_\ell$ paired with the lepton four-momenta; the explicit identity

$$\langle (p_a \wedge p_b) \widetilde{(p_c \wedge p_d)} \rangle_0 = (p_a \cdot p_c)(p_b \cdot p_d) - (p_a \cdot p_d)(p_b \cdot p_c) \quad (6)$$

shows that the invariant projection sits in the grade-zero sector already enumerated by [4, 5]; the six raw covariant components of B_a , however, do *not* reduce to this set and form the natural input channel of the L-GATr-style equivariant networks of refs. [9, 10]. The pair-level azimuthal opening $\Delta\phi_{ij}$ extracts the γ_{12} projection of B_{ij} ; the rapidity difference Δy_{ij} extracts the γ_{03} projection. The kinematic catchment-cone distance $\Delta R_{ij} = \sqrt{\Delta\eta_{ij}^2 + \Delta\phi_{ij}^2}$ is a function of the same projections and is invariant under H_{LHC} but not under full Lorentz. We tabulate the per-grade reading of these spin observables among the grade-2 entries of the 32-row dictionary.

Spin correlations between production and decay, and between different particles in the event, are an independently developed information channel of long standing [32, 34, 35]; the LHC beams are unpolarised, so polarisation enters only through these correlations. The bivector grade carries them as covariant six-component objects rather than as derived hand-crafted scalars, which is the algebraic content of the channel.

¹This embedding is consistent with the algebraic role of ϕ as a coordinate on $\text{SO}(2) \subset H_{\text{LHC}}$ and with the smooth behaviour of equivariant networks across the discontinuity at $\phi = \pm\pi$.

Caveat: event-shape variables and the orthogonal complement. The standard event-shape variables (sphericity S , aplanarity A , thrust T , Fox–Wolfram moments H_ℓ , n -subjettiness τ_N [36], soft-drop mass m_{SD} [37]) are built from a *symmetric* rank-2 tensor, e.g. $S^{ij} = \sum_k p_k^i p_k^j$ for sphericity, and are $SO(3)_{\text{lab}}$ -invariant rather than Lorentz-invariant — they are not antisymmetric blades; they sit in the orthogonal complement of $\bigwedge^2(\mathbb{R}^{1,3})$ inside the space of rank-2 tensors and are therefore not natural elements of the multivector representation. They remain available as auxiliary scalar inputs (entering at the grade-zero projection step) but are not part of the algebraic closure on the kinematic factor; the caveat is reflected as the sphericity-tensor null row of Tab. 5.

What is uniquely new in grade two. Grade two combines the orientation of a decay plane with the rapidity of the relative boost between its participants in a single six-component covariant object (cf. [1, § 5.107]). The sandwich-closure $B \mapsto RB\tilde{R}$ keeps the blade form, which is the algebraic primitive of equivariant networks and is revisited in Sec. 8.

Grade 3 — oriented three-volumes. The grade-three sector of the kinematic factor, $\langle \text{Cl}(1, 3) \rangle_3$, is four-dimensional, spanned by $\gamma_{012}, \gamma_{013}, \gamma_{023}, \gamma_{123}$. The Hodge dual pairs each basis trivector with the corresponding basis four-vector, $(\gamma_{abc})^* = \text{sgn}(\pi_{dabc}) \gamma^d$, where $\text{sgn}(\pi_{dabc})$ is the sign of the permutation taking $(0, 1, 2, 3)$ to (d, a, b, c) so a generic trivector $T \in \langle \mathcal{E} \rangle_3$ is, up to the dual, a covariant four-component object. For three grade-one elements

$$T_{ijk} = p_i \wedge p_j \wedge p_k, \quad T_{ijk} \widetilde{T_{ijk}} = \det G_3 \equiv \det[p_a \cdot p_b]_{a,b \in \{i,j,k\}}, \quad (7)$$

in the convention of Theorem 1. The right-hand side is the Minkowski Gram determinant of the three participating four-momenta and carries its sign through the inner products. A three-subspace of Minkowski space admits at most one time-like direction, and the orthonormal-frame derivation of eq. (B.3) gives $T\tilde{T} = (\det M)^2 \prod_{a=1}^3 \varsigma_a$ with the signs ς_a fixed by the induced signature, so three sub-cases exhaust the possibilities: (i) exactly one time-like direction in the spanned three-plane, induced signature $(1, 2)$, gives $\prod_a \varsigma_a = +1$ and hence $T\tilde{T} = \det G_3 > 0$ — the kinematically dominant case for collider triples; (ii) a fully space-like three-plane, induced signature $(0, 3)$, gives $\prod_a \varsigma_a = -1$ and hence $T\tilde{T} = \det G_3 < 0$ deterministically; (iii) a linearly-dependent (degenerate) triple gives $T\tilde{T} = \det G_3 = 0$ on the boundary.

Physically the trivector is the oriented three-volume in spacetime spanned by the three participating four-momenta, and its boundary $\det G_3 = 0$ is the kinematic edge condition of three linearly dependent momenta — the boundary of a Dalitz-like phase-space region. The Hodge dual T_{ijk}^* is the four-vector orthogonal to the three-plane, equivalently the ϵ -contraction $T^{*\mu} = \epsilon^{\mu\nu\rho\sigma} p_{i\nu} p_{j\rho} p_{k\sigma}$, i.e. the natural covariant counterpart of the $\vec{p}_i \cdot (\vec{p}_j \times \vec{p}_k)$ scalar triple product of the three-spatial reduction.

The classical observable that correlates with grade three is the T-odd / CP-odd triple product of refs. [18, 35] (T-odd in the motion-reversal convention of those references; equivalently P-odd under the covariant-T convention adopted below in the Grade 4 paragraph and in App. C.2.2, where the two conventions are contrasted). In the lab or partonic-CM frame $\vec{p}_i \cdot (\vec{p}_j \times \vec{p}_k) = T_{ijk}^{123}$ is one component of the trivector; the remaining three components $T^{012}, T^{013}, T^{023}$ are sourced by the time-like sector and do not appear in the three-spatial reduction. Under $\text{Spin}^+(1, 3)$ the four components mix as a covariant Lorentz vector through the dual, so a frame-independent equivariant network sees the full four-vector $T_{ijk}^* \in \langle \mathcal{E} \rangle_1$ rather than the single scalar projection T_{ijk}^{123} . The squared three-volume $\det G_3$ reduces to the inner-product sector $\langle \mathcal{E} \rangle_0$ through Theorem 1.

What is uniquely new in grade three is the four-component covariant input rather than the single rest-frame scalar. The squared norm $\det G_3$ is not new — it is captured already by $\langle \mathcal{E} \rangle_0$ via Theorem 1.

Grade 4 — pseudoscalars and the CP-odd sign. The grade-four component $\langle \mathcal{E} \rangle_4 = \lambda I$ is one dimensional, with $I = \gamma_0 \gamma_1 \gamma_2 \gamma_3$ (see the single $\binom{4}{4} = 1$ pseudoscalar slot at the top of Fig. 2), $I^2 = -1$. For four grade-one elements

$$P_{ijkl} = p_i \wedge p_j \wedge p_k \wedge p_l = \lambda_{ijkl} I, \quad \lambda_{ijkl} = \epsilon_{\mu\nu\rho\sigma} p_i^\mu p_j^\nu p_k^\rho p_l^\sigma. \quad (8)$$

The squared magnitude reduces to the four-point Gram determinant through eq. (B.3) for $r = 4$: $\lambda_{ijkl}^2 = -\det[p_a \cdot p_b]_{a,b \in \{i,j,k,l\}}$. The pseudoscalar vanishes identically when fewer than four linearly-independent four-momenta are available, in particular for all final states with three or fewer reconstructed objects after the initial-state sum is included; for ≥ 4 linearly independent participants $\lambda \neq 0$ generically.

Physically λ is the signed four-volume of the parallelepiped spanned by the four momenta in Minkowski spacetime, and its sign is the unique CP-odd one-bit Lorentz-invariant observable *inaccessible* to any classifier that sees only inner products (Theorem 1). Under $\text{Spin}^+(1, 3)$ the pseudoscalar transforms with $\det \Lambda = +1$ and is therefore an invariant of the proper orthochronous group; under the full $O(1, 3)$ the pseudoscalar picks up a sign, $P \rightarrow (\det \Lambda)P$, so the parity transformation P flips $\lambda \rightarrow -\lambda$ (and similarly the covariant time-reversal $\Lambda_T = \text{diag}(-1, +1, +1, +1)$, i.e. $p^0 \rightarrow -p^0$, $\vec{p} \rightarrow +\vec{p}$, flips $\lambda \rightarrow -\lambda$ because each non-zero ϵ -contraction in λ contains exactly one p^0 factor, while C acts trivially on $\text{Cl}(1, 3)$, so λ is P-odd, T-odd, C-even, hence CP-odd). In the Standard Model at tree level the expectation $\langle \text{sgn } \lambda \rangle = 0$ for CP-symmetric configurations; any deviation is a CP-violation signal.

The classical observables correlated with grade four are the sign-asymmetries of refs. [18, 35]. The sign of the Bernreuther observable T_5 reduces, for a chosen four momenta, to $\text{sgn} \langle p_a p_b p_c p_d \rangle_4$; the asymmetries B_3 and $\mathcal{A}_{CP}$ integrate the same one-bit channel over event populations. The signed pseudoscalar has been used as a CP-odd observable since refs. [18, 34, 35]; our contribution is its formulation as a first-class input feature in the GA framework, not its physical existence.

The magnitude $|\lambda| = \sqrt{|\det G_4|}$ reduces to the inner-product sector $\langle \mathcal{E} \rangle_0$ by Theorem 1; the sign $\text{sgn } \lambda$ does not, as the square root of the magnitude loses the orientation information. Together with the four covariant components of grade three (grade-three paragraph above) it gives the full kinematic Lorentz-covariant CP-channel content of the event input layer.

Recovery of the prior input recipes. The recipes of refs. [4, 5] are contained in $\langle \mathcal{E} \rangle_0 \oplus \langle \mathcal{E} \rangle_1$: Lorentz scalars and helicity-frame angular ratios sit in $\langle \mathcal{E} \rangle_0$ (after a rotor sandwich into the chosen frame followed by an inner product), while the lab-frame variables (p_T, η, ϕ) live in the beam-tensored grade-one sector through eq. (5). The equivariant networks of refs. [6, 9, 10] extend the input space to grades ≥ 2 as raw covariant coefficients. The per-variable mapping is given in Tab. 1; the 32-row extension was announced in Sec. 3. The only non-trivial Lorentz-invariant content unlocked by the higher grades is the one-bit sign of the pseudoscalar discussed in the grade-four paragraph above.

4 Operations on the event multivector

The grade decomposition of Sec. 3 fixes the *static* content of the event multivector $\mathcal{E}$ of eq. (4); the *dynamic* content is supplied by a small set of algebraic operations on $\text{Cl}(1, 3)$, each of which carries a direct physical reading on collider four-momenta and a definite role in an equivariant

Table 1: Per-variable mapping of the prior input recipes [4, 5] onto the grade decomposition of the event multivector $\mathcal{E}$ and onto the LHC residual subgroup $H_{\text{LHC}} = \text{SO}^+(1, 1)_z \times \text{SO}(2)_\phi$. Rows 1–14 are standard collider variables, listed here as the union of those recipes; a citation appears only where an entry has a specific origin. “Frame” is the frame in which the variable is defined: Lorentz-invariant (“inv.”), or fixed in the beam, helicity or lab frame. “Inv. subgroup” is the largest group leaving it unchanged: “Y” for full $\text{Spin}^+(1, 3)$, “ H_{LHC} ” for the LHC residual subgroup only, “N” for content not reducible to inner products and masses by Theorem 1, “n.a.” for variables orthogonal to the multivector representation.

#	Variable	Grade	Frame	Inv. subgroup
1	m_i^2	0	inv.	Y
2	$\hat{s}$	0	inv.	Y
3	$\hat{t}$	0	inv.	Y
4	m_{ij}	0	inv.	Y
5	p_T^i	1	beam	H_{LHC}
6	η_i, y_i	1	beam	$\text{SO}(2)_\phi$
7	ϕ_i^a	1	beam	$\text{SO}^+(1, 1)_z$
8	H_T, S_T	1	beam	H_{LHC}
9	$\cos \theta^*$ spin projections [30–32]	0	hel.	Y
10	$\Delta\phi_{ij}, \Delta R_{ij}$	1→0	beam	H_{LHC}
11	C_{ij} spin correlation [34, 35]	0/2	inv.	raw: N; inv. proj.: Y
12	E_T^{miss}	1 _(masked p_z)	beam	H_{LHC} -only
13	b -tag, τ -tag	V_{flav}	n.a.	trivial
14	charge Q_ℓ	V_{flav}	n.a.	trivial
15	T_{ijk}^* trivector dual (this work; cf. [18])	3→1	inv.	raw: $\text{SO}(2)_\phi$ for T^{123} ; covariant: Y
16	$\text{sgn} \langle p_1 p_2 p_3 p_4 \rangle_4$ (this work; cf. [18, 35])	4	inv.	N (CP-odd 1-bit)
17	sphericity S , aplanarity A	n.a.	lab	n.a. (orth. complement)
18	thrust T , Fox–Wolfram H_ℓ [36, 37]	n.a.	lab	n.a. (orth. complement)

^a For neural-network input we recommend the embedding $(\cos \phi^f, \sin \phi^f)$ in place of the raw angle.

network architecture. We collect here the seven operations that are used in the rest of the paper – the geometric, inner, outer and tensor products, the Hodge dual, the rotor sandwich and the bivector exponential, together with the grade-bounded meet and join. The minimal subset used as basic notation throughout is the inner product $p \cdot q$, the reverse $\tilde{X}$ and the grade projection $\langle X \rangle_k$.

Outer product $p \wedge q$. The outer product $p \wedge q = \frac{1}{2}(pq - qp)$ of two grade-one elements is the oriented Minkowski plane spanned by the two four-momenta; it is closed in the grade-two sector $\langle \text{Cl}(1, 3) \rangle_2$ of dimension six and inherits its squared magnitude from the Cayley–Menger Gram-determinant identity of Theorem 1, so that $|p \wedge q|^2 = (p \cdot q)^2 - p^2 q^2$ in the convention of eq. (B.2). Physically the two-blade $p_\ell \wedge p_\nu$ encodes the leptonic decay plane of a W boson and $(p_\ell + p_\nu) \wedge p_b$ the corresponding plane of the parent top; their bivector products supply the Bernreuther spin-correlation projector $C_{ij}^{\text{Bernreuther}}$ of refs. [34, 35] (grade-2 entries of the dictionary). In the equivariant-network setting the raw two-blade is supplied to the network as a covariant grade-two input in the sense of ref. [10]; its scalar magnitude is the polynomial already accessible from $\langle \mathcal{E} \rangle_0$ through Theorem 1.

Geometric product $pq = p \cdot q + p \wedge q$. The geometric product reunifies the symmetric and antisymmetric parts of the inner and outer products into the single bilinear operation that closes the algebra at the sixteen-dimensional level of eq. (3). It is the universal Lorentz-equivariant primitive of the geometric-algebra-transformer line of refs. [8–10]: every equivariant bilinear map on multivector inputs is generated by grade projections of the geometric

product together with its pseudoscalar-dressed counterpart $\langle xIy \rangle_k$ (the join), since I is itself $\text{Spin}^+(1, 3)$ -invariant. The closure under the geometric product is what makes $\text{Cl}(1, 3)$, rather than the underlying Minkowski vector space, the natural carrier for an equivariant input layer.

Hodge dual $X^* \equiv XI^{-1}$. The Hodge dual pairs grade k with grade $4 - k$ (Fig. 1, dashed arcs) and supplies the covariant four-vector representation of the trivector dual T_{ijk}^* (grade-3 entries of the dictionary), the natural container for the Atwood–Soni-style triple-product CP-odd features [18]. The dual appears here in two places: in the construction of the meet (eq. (9), Sec. 4), and in the identification of the pseudoscalar coefficient $\langle p_1 p_2 p_3 p_4 \rangle_4$ as the unique CP-odd one-bit input feature of $\mathcal{E}$. The familiar electromagnetic instance $\star F = FI^{-1}$ is the same pairing; sign-convention details vs. the right-multiplication form $\star F = FI$ of ref. [1] are in the notation card. Self-duality on grade-two, the Weyl-spinor parallel and fermionic extensions are deferred to Sec. 9.

Rotor sandwich $X \mapsto RX\tilde{R}$. The proper orthochronous Lorentz group acts on a multivector through the sandwich product of eq. (2) with R in the spin cover $\text{Spin}^+(1, 3) \simeq \text{SL}(2, \mathbb{C})$, generated by a bivector $B \in \langle \text{Cl}(1, 3) \rangle_2$ via $R = \exp(B/2)$. The sandwich is grade-preserving and realises any Lorentz transformation as a single algebraic operation without reference to a 4×4 matrix on coordinates: the laboratory-to-helicity transition R_{hel} , the laboratory-to-Collins–Soper transition R_{CS} , and the Mahlon–Parke optimal-spin-axis rotation [32] are each implemented by one rotor that acts uniformly on every grade of $\mathcal{E}$. This is the operation around which all current geometric-algebra equivariant networks [8–10] and Lorentz-equivariant networks [6, 7] are built; the input recipe of the present paper is engineered to feed those networks with the smallest set of features that contains everything they need.

Tensor product $\text{Cl}(1, 3) \otimes V_{\text{flav}}$. The construction of eq. (4) attaches the discrete object-type label of each detector object as a Lorentz-trivial tensor factor V_{flav} in the basis of eq. (14). We are explicit that this is not an algebraic operation internal to $\text{Cl}(1, 3)$: it is the formalisation of the auxiliary scalar channel introduced as a per-token feature in the geometric-algebra transformers of ref. [10]. The tensor product factorises the input as a kinematic Clifford factor times an object-type factor (Fig. 3), which is the interface through which a single architecture handles all detector object classes uniformly while keeping the Lorentz action of eq. (10) confined to the $\text{Cl}(1, 3)$ side. Two algebraically richer alternatives – an extended Clifford algebra $\text{Cl}(1, 3+n_f)$ (Variant B) and a multi-particle spacetime algebra [1, 2] (Variant C) – are analysed and rejected in Sec. 6, on grounds of dimension and of equivariance under a symmetry the data does not possess.

Meet and join. The join and meet of two blades are

$$A \cup B \equiv A \wedge B, \quad A \cap B \equiv (A^* \wedge B^*)^{-*}, \quad (9)$$

the join raising the grade and the meet lowering it through the double dual. Geometrically, the join is the smallest blade whose subspace contains those of both operands: two linearly independent four-momenta join into the bivector $p_i \wedge p_j$ spanning their decay plane, and a four-momentum joined with a decay-plane bivector yields the trivector spanning the three-volume of the corresponding top decay. The meet is its dual: the largest blade whose subspace lies in both, i.e. their geometric intersection. In $\text{Cl}(1, 3)$ the I -dual meet of two distinct decay-plane bivectors is a *scalar* — the grade bound collapses the intersection information, as quantified below — while in the conformal extension of Sec. 9 (eq. (17)) two mass-shell blades meet in

a genuine lower-grade blade encoding the kinematic intersection of the corresponding hyperboloids. The Hodge dual in the meet formula realises intersection as a join of orthogonal complements: dualising A and B into their orthogonal complements, joining those, and dualising back recovers the largest blade contained in both originals. Both operations are *grade-bounded* in $\text{Cl}(1, 3)$, by which we mean that the sum of the grades of the operands is constrained by the maximum grade 4 of the algebra, so that generic higher-grade combinations collapse: the meet of two distinct two-blades reduces to a Lorentz-invariant scalar already covered by Theorem 1, and the join of three generic two-blades vanishes by dimension count. The richer multi-resonance arena is the conformal extension $\text{Cl}(2, 4)$, in which the grade bound is lifted: this construction is given as a representational specification in Sec. 9. The outermorphism property $f(A \wedge B) = f(A) \wedge f(B)$ of any linear map f [2] ensures that the bivector channel of multi-resonance attention layers preserves grade structure under any per-token Lorentz transform, which is the architectural prerequisite for stacking equivariant heads on multi-blade inputs. Eq. (9) is the operation the pairing tokens of Sec. 7 apply to the two candidate top decays: the meet $T_{\text{had}}^{(a)} \cap T_{\text{lep}}^{(a)}$ of the two candidate top-decay trivectors, a grade-two covariant whose invariant norm indicates coplanarity of the two decay volumes in the multi-resonance reconstruction problem. Being a function of the Gram matrix, its invariant norm is redundant by Theorem 1; Sec. 7.1 measures what supplying it explicitly is worth on the tWb benchmark. The conformal embedding of Sec. 9 provides genuine grade-raising meets and direction-valued geometric primitives for resonance-mass shells.

Bivector exponential $\exp(B/2)$. The map $B \mapsto R = \exp(B/2)$, with B in the six-dimensional bivector sector $\langle \text{Cl}(1, 3) \rangle_2$ and R in the spin cover $\text{Spin}^+(1, 3)$, supplies the smooth manifold structure on which a learnable rotor parameter lives in equivariant networks [8, 10]. The bivector sign rule $\text{sgn}(B^2)$ separates the rotation generators ($B^2 < 0$ in our signature) from the boost generators ($B^2 > 0$): for instance, $B = \alpha \gamma_{03}$ ($B^2 > 0$) exponentiates to a longitudinal boost of rapidity α , while $B = \phi \gamma_{12}$ ($B^2 < 0$) exponentiates to an azimuthal rotation by ϕ . One bivector parameter spans the full set of Lorentz transformations without reference to the rotation/boost split required by tensor formulations.

5 Symmetries and equivariance

Continuous spacetime symmetries. The full symmetry of the kinematic part of $\mathcal{E}$ is the spin cover $\text{Spin}^+(1, 3) \simeq \text{SL}(2, \mathbb{C})$ of the proper orthochronous Lorentz group, acting by the sandwich product

$$X \longmapsto RX\tilde{R}, \quad R \in \text{Spin}^+(1, 3). \quad (10)$$

Equation (10) is grade-preserving and acts as the fundamental representation on grade one, the adjoint on grade two, and the corresponding higher representations on grades three and four. The $\text{Spin}^+(1, 3)$ -invariants of $\mathcal{E}$ are the grade-zero coefficients enumerated in Sec. 3 together with the grade-four coefficient, which is invariant because $\det \Lambda = +1$ on the proper orthochronous group; grades one to three transform equivariantly, which is the form in which they enter the networks of refs. [6–10] (the full list of continuous spacetime symmetries acting on $\mathcal{E}$, with their generators and enforcement strategies, is collected in the symmetry inventory of App. C.2 (Tab. 6); the per-bucket tables there are cited by number below).

The LHC subgroup. Choosing a beam axis $\hat{n}_z$ and a transverse plane breaks $\text{Spin}^+(1, 3)$ to the longitudinal-boost $\times$ azimuthal-rotation subgroup

$$H_{\text{LHC}} \equiv \text{SO}^+(1, 1)_z \times \text{SO}(2)_\phi \subset \text{Spin}^+(1, 3) \quad (\text{via the rotor double cover of eq. (2)}), \quad (11)$$

the same subgroup that acts diagonally on the LHC beam-axis parameterisation (p_T, y, ϕ, M_T) of eq. (5). Missing transverse momentum from one or more neutrinos restricts the equivariance further within H_{LHC} , since the longitudinal momentum of the neutrino system is unobserved. Both restrictions are common to all collider applications of equivariant networks (see e.g. ref. [10]); neither is a property of the present algebraic representation.

Algebraic 2+2 split. The algebraic split adapted to H_{LHC} ,

$$\text{Cl}(1, 3) \cong \text{Cl}(1, 1)_z \hat{\otimes} \text{Cl}(0, 2)_\perp, \quad (12)$$

splits the algebra along the longitudinal/transverse axis. The first factor carries the longitudinal boost subgroup $\text{Spin}^+(1, 1)$, the second factor the azimuthal rotation subgroup $\text{Spin}^+(0, 2)$, and the graded tensor symbol $\hat{\otimes}$ tracks the $\mathbb{Z}_2$ -graded anticommutation between them. The (y, M_T) content of an event lives in $\text{Cl}(1, 1)_z$ and the (p_T, ϕ) content in $\text{Cl}(0, 2)_\perp$, matching the standard decomposition of LHC kinematic variables. Eq. (12) is used below to read off the equivariance content of each symmetry bucket.

Which fixed blades are H_{LHC} -invariant. The split makes it immediate which constant algebraic objects a network may carry as fixed reference tokens without breaking H_{LHC} . The two generators are the longitudinal boost $\gamma_{03} \equiv \gamma_0 \gamma_3$ and the azimuthal rotation γ_{12} , and they act on a constant blade B by the commutator. Now γ_{03} commutes with itself, and it commutes with γ_{12} because the two blades share no index, so

$$[\gamma_{03}, \gamma_{03}] = [\gamma_{12}, \gamma_{03}] = 0, \quad (13)$$

and the beam bivector γ_{03} — equivalently the (t, z) plane, the first factor of eq. (12) — is invariant under the whole of H_{LHC} . The individual vectors γ_0 and γ_3 are not: each is rotated into the other by the longitudinal boost, $[\gamma_{03}, \gamma_0] = -2\gamma_3$ and $[\gamma_{03}, \gamma_3] = -2\gamma_0$. A network supplied with γ_0 and γ_3 separately therefore has the ingredients of the H_{LHC} invariants — $E_i = \langle T_i \gamma_0 \rangle_0$ and $p_{z,i} = -\langle T_i \gamma_3 \rangle_0$, from which p_T follows — but is not itself H_{LHC} -equivariant, whereas one supplied with γ_{03} is, by construction. This distinction is not decorative: Sec. 7.1 measures the two choices against each other, and they differ by 0.00742 ± 0.00047 in AUC at identical parameter count.

Discrete spacetime symmetries. Parity P , time reversal T , and the combined CP act as outer automorphisms of $\text{Cl}(1, 3)$. Parity sends $\gamma_0 \mapsto +\gamma_0$, $\gamma_i \mapsto -\gamma_i$ and is implemented by the sandwich $X \mapsto \gamma_0 X \gamma_0^{-1}$. Time reversal flips γ_0 and is implemented as an antilinear involution. Charge conjugation C does not act on $\text{Cl}(1, 3)$; it acts non-trivially on the object-type space V_{flav} by exchanging particle/antiparticle labels (see Sec. 6). The combined CP is therefore an action on the full $\text{Cl}(1, 3) \otimes V_{\text{flav}}$ in which the spacetime piece is the parity sandwich and the object-type piece is the involution induced by charge conjugation. The full list of discrete spacetime symmetries (P , T , CP , CPT) and their action on the grades of $\mathcal{E}$ is given in Tab. 7.

Permutational symmetry. Reconstructed objects of the same type are physically exchangeable: swapping two electrons, or two light jets, leaves the event invariant. This is realised on $\mathcal{E}$ as the permutational symmetry S_{n_a} for each object-type label a in the basis of V_{flav} , acting as a relabelling of the per-object copies in the multi-particle algebra of Sec. 2. In a network architecture this symmetry is enforced by attention with shared per-token weights rather than by a symmetric pooling operation, in line with all current set-attention architectures [10].

Internal and gauge symmetries. The electromagnetic gauge $U(1)_{\text{em}}$ is represented in the feature space through a charge-bit embedding on V_{flav} (see Sec. 6; row C.1), providing the network with explicit charge information as a design feature rather than as an inferred property. The electroweak gauge $SU(2)_L \times U(1)_Y$, the colour gauge $SU(3)_C$, and the flavour-mixing matrices CKM and PMNS are not encoded in the feature space; they are handled at the amplitude level in the Monte-Carlo event generation (rows C.2–C.5). Their role is implicit through the structure of the generated events, and no hard architectural constraint enforces them on $\mathcal{E}$. Lepton-flavour universality, realised on V_{flav} as an S_3 permutation among e, μ, τ tokens of the same lepton-type label, can be implemented as an optional weight-tying mechanism on the corresponding feature channels; the breaking by the charged-lepton masses is documented rather than constrained, and the full implementation is deferred to the foundation-model program of Sec. 8. The complete internal/permutational inventory is collected in Tab. 8.

Approximate, soft, and CP-channel symmetries. Beyond strict equivariance, several softer symmetries shape the architecture (Tab. 9). Dilatation $SO(1, 1)_D$, representing a global scaling of all four-momenta, is not a symmetry of the SM Lagrangian but is approximately preserved by detector-level kinematics; it is addressed through layer normalisation, treated as a soft architectural choice rather than as a hard symmetry constraint. The pseudoscalar sign $\mathbb{Z}_2$ observable $\text{sgn} \langle p_1 p_2 p_3 p_4 \rangle_4$ of Sec. 3 (App. C.2.4 row G.1) enters the demonstration of Sec. 7 as a single CP-odd one-bit input feature; on tree-level Standard-Model configurations of the present example it acts as a null test, a property exploited in the demonstration but not enforced as a symmetry of the network. The absence of a learnable positional encoding on the per-object tokens (row F.2) is an absent-by-design architectural choice that preserves S_{n_a} invariance. Other approximate symmetries occasionally invoked at colliders — the running-coupling $SO(1, 1)$ of the top- p_T spectrum, the approximate $SU(N_f)_L \times SU(N_f)_R$ chiral symmetry of QCD with massless quarks (row E.1), the custodial $SU(2)_V$ symmetry of the EW Higgs sector (row E.3), heavy-quark spin-flavour symmetry, and BSM CP and EDM channels — are out of scope for the present feature-space design and are deferred to the foundation-model program of Sec. 8.

Symmetry inventory. The complete inventory of 30 symmetries acting on $\mathcal{E}$ is collected in App. C.2, organised into four functional buckets (continuous spacetime, discrete spacetime, internal and permutational, approximate / ML-architectural / CP) and classified by enforcement strategy (*strict, soft, embedding, conditional, automatic, out-of-scope*).

6 Object-type space V_{flav}

The discrete reconstructed information — charged-lepton flavour, the electric-charge bit, b - and τ -tag bits, the missing transverse momentum — is a Lorentz-trivial complement to the kinematic Clifford factor of the event multivector. We collect it in a finite-dimensional real vector space V_{flav} that is tensored to $\text{Cl}(1, 3)$ and on which $\text{Spin}^+(1, 3)$ acts as the identity. The construction follows the auxiliary-scalar pattern of the equivariant transformer [10] (Variant A in the taxonomy below); algebraically richer alternatives are recorded as discussion items and deferred to subsequent work. Discrete symmetries acting on V_{flav} — the charge-conjugation involution C , the parity action P inherited from the kinematic factor, and the per-flavour permutational S_n on identical-token buckets — are inventoried in Sec. 5 and App. C.2.3.

The event multivector $\mathcal{E}$ itself was defined in Sec. 2.2; this section constructs the object-type factor V_{flav} it requires. The order of the slots in eq. (4) is immaterial: a network operating on $\mathcal{E}$ realises the permutational symmetry of identical detector objects through set-attention rather than through the algebraic sum. The action of $\text{Spin}^+(1, 3)$ on $\mathcal{E}$ acts only on the $\text{Cl}(1, 3)$

factor through the sandwich product of eq. (10), leaving the V_{flav} component invariant. This is the statement that a global change of inertial frame does not change particle identity (an electron remains an electron in every Lorentz frame): the sandwich boosts every p_i and, by the outermorphism property of Sec. 4, every multi-particle blade $p_i \wedge p_j \wedge \dots$ generated from them by the same global rotor R , but it does not touch the object-type labels $|f_i\rangle$. The multi-particle mixing of momenta across different flavours — the leptonic- W plane $p_\ell \wedge p_\nu$ combining an e/μ with E_T^{miss} , the top-decay trivector $p_\ell \wedge p_\nu \wedge p_b$ combining three different object types, and so on — is produced by the algebra operations of Sec. 4, *not* by the Lorentz action. The discrete symmetries C , P , T , which do permute object types and charges, act non-trivially on V_{flav} and are inventoried separately in Sec. 5.

Object-type basis. For the LHC final states of interest in the present paper – the single-lepton-plus-jets signature of the tWb demonstration and the broader class of t -quark physics signatures of the HiGEN (Hierarchical Geometric Event Network) program – we use the basis

$$\mathbb{F} = \{e^+, e^-, \mu^+, \mu^-, \tau_h^+, \tau_h^-, j, bj, \gamma, E_T^{\text{miss}}\}, \quad \dim V_{\text{flav}} = 10. \quad (14)$$

This particular basis is an *instance*, not a property of the construction. Nothing in eqs. (4)–(10) fixes $\mathbb{F}$ or its dimension: V_{flav} is a parameter of the reconstruction vocabulary, chosen here to cover the standard reconstructed object types of ATLAS and CMS analyses of top-quark physics at the operating point of refs. [11–13], which gives $\dim V_{\text{flav}} = 10$. Additional tags (c -tag, boosted $W/Z/H/t$ substructure tags, forward-tagged jets) enlarge $\mathbb{F}$ without touching anything else in the formalism.

Two consequences should be stated plainly, since the reconstruction level and the truth level do not coincide. The basis of eq. (14) is a *reconstruction-level* vocabulary: it carries missing transverse energy rather than a neutrino, and it does not separate quark from antiquark, both light jets mapping to j and both b -jets to bj . The parton-level demonstration of Sec. 7 therefore does not use it. That demonstration works at truth level with the six-element basis $\{\mu^-, \bar{\nu}_\mu, u, \bar{d}, b, \bar{b}\}$, in which the neutrino is an explicit four-momentum and quark charge is known, so the missing-energy encoding of eq. (15) is not exercised there and the flavour-tagging caveats below do not apply to it. We flag the difference here because the two bases have different dimensions and a reader tracking $\dim V_{\text{flav}}$ through the paper would otherwise find them inconsistent.

Variant choice. We adopt Variant A (the tensor product $\text{Cl}(1, 3) \otimes V_{\text{flav}}$ of eq. (4)) throughout the paper: the Lorentz factor commutes with the discrete object-type relabellings, the algebraic content of $\mathcal{E} = p \otimes |f\rangle$ is the tensor product, and the discrete labels $|f\rangle$ live in a fixed real vector space rather than in an extended Clifford algebra, in line with the auxiliary-scalar channel of ref. [10]. Variant B — the extended Clifford algebra $\text{Cl}(1, 3 + n_f)$, in which n_f extra basis vectors carry the object-type labels and the algebraic charge-conjugation automorphism acts non-trivially on the flavour generators — is rejected here rather than deferred, and it is worth saying why explicitly, since it is the only alternative with a genuine algebraic advantage.

That advantage is real but narrow: C becomes an inner automorphism of the algebra instead of an involution on a separate factor. The costs are three. First, dimension. The extended algebra has 2^{4+n_f} basis elements per token, which for the reconstruction vocabulary of eq. (14) ($n_f = 10$) is 16 384 against $\dim(\text{Cl}(1, 3) \otimes V_{\text{flav}}) = 16 \times 10 = 160$ — a factor 102, and the comparison is against the tensor product actually used, not against $\text{Cl}(1, 3)$ alone. Second, and more seriously, the enlarged spin group $\text{Spin}(1, 3 + n_f)$ contains transformations mixing space-time with object-type directions, for which there is no physical symmetry to be equivariant to; equivariance under the full group would be equivariance under a symmetry the data does not have. Third, we are aware of no practical off-the-shelf equivariant layer implementation at

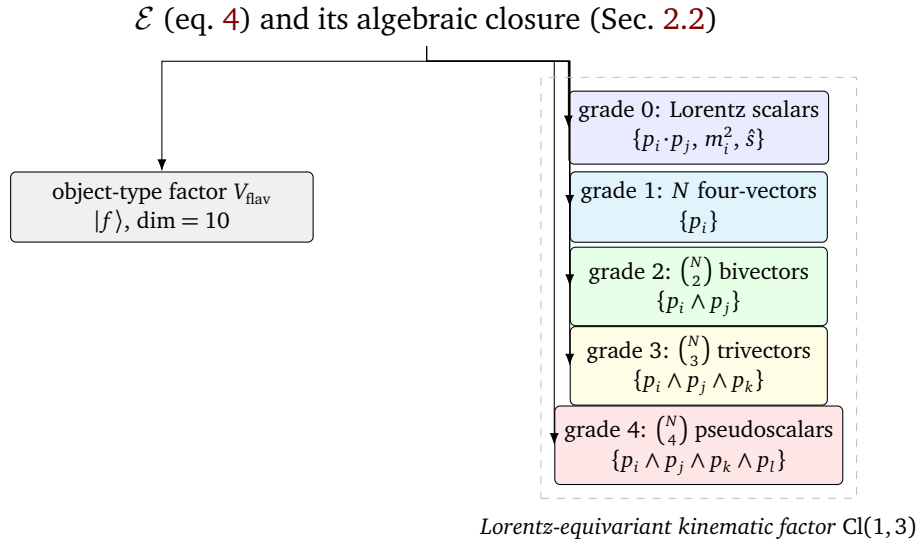


Figure 3: The event multivector $\mathcal{E}$ as a hierarchical decomposition into the kinematic Clifford factor $\text{Cl}(1, 3)$ and the Lorentz-trivial object-type factor V_{flav} . The per-grade counts on the right are for an event with N final-state detector objects (the parton-level tWb demonstration of Sec. 7 uses $N = 6$; reconstructed-level single-lepton-plus-jets samples typically extend to $N \lesssim 8$ once ISR jets are included). The object-type factor on the left carries one entry per detector token from the basis of eq. (14). The per-grade physical reading of the kinematic factor is developed in Sec. 3. Grades ≥ 2 arise from the closure of the grade-one input (Sec. 2.2), not from eq. (4) itself; their type labels live in tensor powers of V_{flav} (Tab. 8).

this dimensionality, so the construction could not be trained today without writing them. We therefore keep the tensor product, in which the Lorentz action stays confined to the $\text{Cl}(1, 3)$ factor by eq. (10), and record Variant B as closed rather than open. Variants C (the multi-particle Clifford bundle $\text{Cl}(1, 3)^{\otimes N}$ of refs. [1, 2]) and D (the $\mathbb{Z}_2$ -graded extension implicit in the algebraic action of C) are not implemented in the present paper. The C-equivariance of the V_{flav} -tensored representation is therefore not enforced as an algebraic symmetry; it is broken by the choice of basis in eq. (14), in which the particle/antiparticle labels are independent unit vectors rather than the ± 1 eigenstates of an algebraic charge-conjugation involution. The cost of this choice is a one-bit asymmetry in the network's response to charge-conjugated events; the benefit is a direct compatibility with the ATLAS/CMS object reconstruction conventions and with the scalar channel of refs. [9, 10].

The missing-energy pseudo-object. The reconstructed missing transverse momentum is the two-dimensional vector $\vec{p}_T^{\text{miss}} \equiv -\sum_{i \in \text{vis}} \vec{p}_T^i$ of visible-transverse-momentum imbalance, with components $(p_x^{\text{miss}}, p_y^{\text{miss}})$ and scalar magnitude $E_T^{\text{miss}} \equiv |\vec{p}_T^{\text{miss}}|$ (the missing transverse energy proper). The pair $(\vec{p}_T^{\text{miss}}, E_T^{\text{miss}})$ is treated as a pseudo-object with grade-one component

$$p_{E_T^{\text{miss}}} = (0, p_x^{\text{miss}}, p_y^{\text{miss}}, 0), \quad f_{E_T^{\text{miss}}} = |E_T^{\text{miss}}\rangle, \quad (15)$$

together with a binary measurement mask $m_{E_T^{\text{miss}}}^{\text{meas}} = (0, 1, 1, 0)$ in V_{flav} flagging the energy and longitudinal slots as unmeasured. The mask is propagated through downstream layers so that the apparent space-like value $p_{E_T^{\text{miss}}}^2 < 0$ is never used as a physical quantity. The Lorentz transformation rule of $p_{E_T^{\text{miss}}}$ is the restriction of the rule for a generic four-momentum to the LHC subgroup H_{LHC} of eq. (11), since the longitudinal boost component is unmeasured. We

adopt the partial four-momentum of eq. (15) for input-layer uniformity; three alternatives are summarised in Tab. 2.

Table 2: Candidate algebraic encodings of E_T^{miss} . Adopted choice: partial four-momentum (row 1).

Encoding	Grade	Preserves	Cost
Partial four-momentum (eq. (15))	1 (masked E, p_z)	per-token uniformity	explicit binary mask
Transverse two-blade $p_T^{\text{miss}} \wedge \hat{n}_z$	2	$\text{SO}(2)_\phi$ orbit	breaks per-token uniformity
$p_z(\nu)$ from W -mass [11, 12]	1 (full)	full Lorentz orbit	quadratic; single- ν
2+2 split $\text{Cl}(1, 1) \otimes \text{Cl}(0, 2)$	basis change	LHC-natural longitudinal/transverse symmetry	non-trivial rewrite

Discrete symmetries on V_{flav} . The discrete symmetries acting on the object-type factor are catalogued in App. C.2.3: the charge-conjugation involution C acting on V_{flav} as a $\mathbb{Z}_2$ exchanging particle/antiparticle labels (row C.1); the per-flavour permutation $\prod_a S_{n_a}$ realised by set-attention (rows D.1–D.5); and the lepton-flavour-universality S_3 permutation acting among e, μ, τ tokens of the same lepton-type label as a soft weight-tying option. The combined CP acts on $\text{Cl}(1, 3) \otimes V_{\text{flav}}$ as the parity sandwich on the kinematic factor times the C -involution on the object-type factor. We do not enlarge V_{flav} to a $\mathbb{Z}_2$ -graded extended Clifford algebra here; the algebraically richer Variant B alternative is analysed above and rejected there, not deferred.

Impact-parameter caveat. The b -tag information is reduced to the binary slot $|bj\rangle \in \mathbb{F}$; the underlying impact-parameter and secondary-vertex kinematics that enter at reconstruction are not encoded in the input layer, and a full vertex-level treatment is deferred to Sec. 8.

Spin-state caveat. The event multivector $\mathcal{E}$ encodes 4-momenta and reconstructed object-type flags only; fermion spin states are not represented as Dirac spinors $\Psi \in \text{Cl}(1, 3)$, and spin information enters implicitly via the second moment $\mathcal{E} \otimes \mathcal{E}$ (decay-plane bivectors and Bernreuther correlations of Sec. 3).

7 Resonance-topology separation in $pp \rightarrow tWb$

In this section we apply the algebraic representation of the preceding sections to a concrete physical example, the parton-level separation of single- and double-resonant contributions to the $pp \rightarrow tWb$ final state, and compare a compact geometric-algebra network on multivector input (GA) with the established high-level reference network of ref. [11] (REF). Realistic detector effects, calibration and a data fit are out of scope of this comparison and are subjects of dedicated separate studies.

Physics target. The process $pp \rightarrow tWb$ is the lowest-multiplicity hadron-collider final state in which a single Breit–Wigner top resonance and a pair of them coexist as components of the same matrix element. The associated single-top channel $pp \rightarrow tW^-\bar{b}$ contributes one resonant top through $t \rightarrow W^+b$, while the doubly-resonant $t\bar{t}$ component of the same six-particle final state contributes two through $t \rightarrow W^+b$ and $\bar{t} \rightarrow W^-\bar{b}$; the two topologies are two diagram

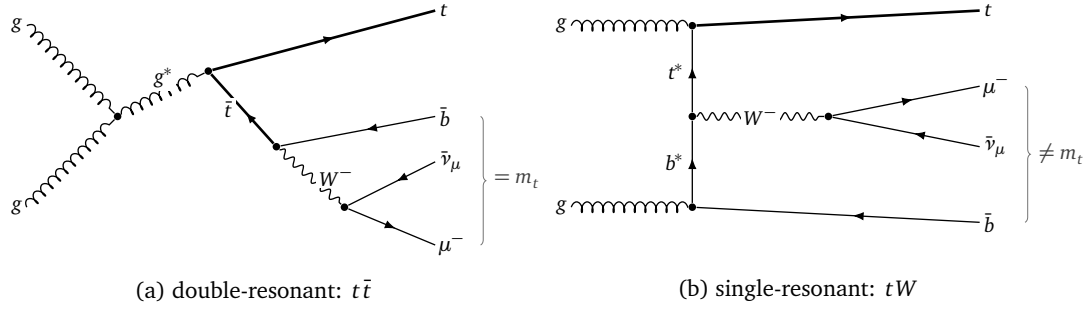


Figure 4: Two representative tree-level diagrams contributing to the $gg \rightarrow t\bar{b}\mu^-\bar{\nu}_\mu$ final state of the matrix-element set of refs. [12, 13]. The braces on the right indicate the invariant mass of the $\bar{b}\mu^-\bar{\nu}_\mu$ trio — the observable that distinguishes the two topologies. (a) **Double-resonant** ($t\bar{t}$): $\bar{b}$, μ^- and $\bar{\nu}_\mu$ originate from the same $\bar{t}$ -line, so $m_{\bar{b}\mu^-\bar{\nu}_\mu} = m_t$ on the Breit-Wigner pole. (b) **Single-resonant** (tW): the $\bar{b}$ comes from a $g \rightarrow b\bar{b}$ splitting rather than from a top decay, so the trio does *not* share a common top resonance and $m_{\bar{b}\mu^-\bar{\nu}_\mu} \neq m_t$. The two components coexist as gauge-invariant pieces of the same matrix element [12, 13].

classes of the same matrix element. Differential distributions are sensitive to spin correlations between production and decay and between distinct particles within the same event, a well-developed phenomenological block [30, 32, 35]; the present demonstration is intended to expose how the algebraic representation handles the resonance topologies grade by grade, not to claim a new measurement of them. The same separation problem arises, with more resonances and a larger combinatorial burden, in the modelling of triple-top production [38] and of four-top final states, where diagram classes carrying three and four simultaneous top resonances share a final state with the lower-multiplicity classes. The relevance of this tWb system has been sharpened by recent observations of a cross-section enhancement near the $t\bar{t}$ production threshold by CMS [39] and ATLAS [40, 41], consistent with the formation of a colour-singlet quasi-bound toponium state with cross sections of $8.8^{+1.2}_{-1.4}$ pb and $9.3^{+1.4}_{-1.3}$ pb respectively. The modelling of the same gauge-invariant tWb matrix element used here — and in particular the treatment of the $t\bar{t}/tW$ overlap that the two truth classes of the next paragraph isolate — enters those measurements as one of the leading sources of systematic uncertainty, so an input representation that exposes the resonance topology directly addresses an open modelling issue in top physics. Representative diagrams of the two resonance topologies are shown in Fig. 4(a),(b).

Truth definition of the two classes. The two classes are defined at the matrix-element level through a diagram-removal scheme [11, 12]. Class A (single-resonant) is generated keeping only the electroweak tree-level tWb diagrams that produce a single top resonance through $t \rightarrow W^+b$. Class B (double-resonant) is generated keeping the $t\bar{t}$ diagrams that produce two top resonances through $t \rightarrow W^+b$ and $\bar{t} \rightarrow W^-\bar{b}$. A third sample is generated with the full gauge-invariant matrix element, including both classes and their interference; it is not used for training and serves only as a cross-check on the inclusive event population (third curve of the discriminant plot below). Truth labels are taken from the parton-level record of the matrix-element generator.

Generator and event selection. Events are generated at $\sqrt{s} = 14$ TeV with the matrix-element generator COMPHEP 4.5 [42] for the six-particle final state $gg \rightarrow \mu^-\bar{\nu}_\mu u\bar{d}b\bar{b}$, in which the leptonic decay $\bar{t} \rightarrow \bar{b}W^- \rightarrow \bar{b}\mu^-\bar{\nu}_\mu$ and the hadronic decay $t \rightarrow bW^+ \rightarrow bu\bar{d}$ enter

together with the associated single-top production $gg \rightarrow tW^- \bar{b}$ (t hadronic, $W^- \rightarrow \mu^- \bar{\nu}_\mu$) as diagram classes of the same matrix element for this fixed final state. The single- and double-resonant contributions and their interference are produced as components of the same matrix element. The demonstration is performed at the parton level: neither parton shower nor detector simulation is applied, so that the test isolates the effect of the algebraic input representation from shower and detector physics. The matrix-element scheme, parton distribution, factorisation and renormalisation scales, and the event selection $p_T^{j_4} > 10$ GeV reproduce the setup of the reference network of ref. [11].

Reference network. The reference network REF of ref. [11] is a fully-connected feed-forward neural network on ~ 75 high-level Lorentz-invariant features, combining the recipes of refs. [4, 5]: logarithms of pairwise dot products of the reconstructed four-momenta $\log p_i \cdot p_j$ (which are positive on the kinematics of interest and equivalent to logarithms of two-body invariant masses for light final-state partons), logarithms of transverse momenta $\log p_T$, logarithms of the reconstructed top and W masses and of $\hat{s}$, E_T^{miss} , H_T , and a complete set of pseudorapidities and helicity-frame cosines that enter the network without a logarithm. Three hidden layers of five hundred units each, with L_2 -regularisation and dropout, are trained on 10^5 tW and 10^5 $t\bar{t}$ parton-level events with the cut $p_T^{j_4} > 10$ GeV; binary cross-entropy with inverse-class-frequency weighting and full-batch Adam are used. The published checkpoint contains 5.4×10^5 trainable parameters. Training and a complete list of input variables are documented in ref. [11]; we reuse the published checkpoint without retraining.

Geometric-algebra network. A compact L-GATr variant (informally, GATr-lite) is used: it is a compact multivector-equivariant transformer in the family of ref. [10] that matches REF at the parton level on this benchmark, so the comparison probes the inductive bias of the algebraic input layer rather than the network capacity. The geometric-algebra network GA ingests the same events as multivectors of $\text{Cl}(1, 3) \otimes V_{\text{flav}}$. Each of the six final-state partons of the generator setup described above is represented as one token of the form $T_i = \langle T_i \rangle_0 \oplus \langle T_i \rangle_1$, where $\langle T_i \rangle_1 = p_i^\mu \gamma_\mu$ is the four-momentum and $\langle T_i \rangle_0$ carries a one-hot encoding of the object type (μ^- , $\bar{\nu}_\mu$, u , $\bar{d}$, b , $\bar{b}$), together with the electric charge and b -tag flags as scalar channels. The network has three equivariant blocks, each composed of (i) a per-grade *equivariant linear* layer mixing channels within each grade [10], (ii) a *geometric-product* layer that contracts pairs of multivector channels and projects the result back through grade-aware linear mixing, so that bivectors $p_i \wedge p_j$ and trivectors $p_i \wedge p_j \wedge p_k$ enter the network as covariant objects of the corresponding grades, and (iii) a *Lorentz-invariant attention* head whose attention scores are scalar inner products $\langle T_i, T_j \rangle_0$ of multivector keys and queries. Equivariance under $\text{Spin}^+(1, 3)$ holds by construction: every operation either projects to grade-zero scalars (preserving invariance) or acts on multivector channels by per-token rotor sandwich (preserving covariance). The readout extracts the grade-zero coefficients of the final tokens, applies a permutation-invariant mean pool over the six tokens and feeds a small dense head to the binary discriminant. With the candidate-pairing tokens the network has 151,489 trainable parameters against 539,501 for REF, a factor 3.6 below it, so that the comparison probes inductive bias rather than capacity. Training uses AdamW with a cosine learning-rate schedule (initial $\lambda = 10^{-3}$, weight decay 10^{-4}), mini-batch size 512, gradient-norm clipping at 1.0, and binary cross-entropy loss; GA is trained on the full parton-level Monte-Carlo statistics passing the same selection $p_T^{j_4} > 10$ GeV as REF ($\sim 270\text{k}$ tW and $\sim 10\text{M}$ $t\bar{t}$ events before the held-back split; 9.3M enter training) with a balanced random sampler that draws $\sim 500\text{k}$ events per epoch. Training is run for 50 epochs, a few hours per configuration on a single consumer-class GPU. Every AUC is taken at the last epoch, so that no model is selected by the quantity being reported.

Event-level pairing tokens. Both tW and $t\bar{t}$ share the same six-particle final state (μ^- , $\bar{\nu}_\mu$, two light jets j_1, j_2 , and two b -tagged jets b_1, b_2). The discrimination between single- and double-resonant topologies is, at detector level, inseparable from the combinatorial assignment of the two b -tagged jets to the candidate top-quark resonances — the classical reconstruction problem of associated tWb production. At the parton level used here the object-type labels distinguish b from $\bar{b}$ and thus fix the assignment; the pairing tokens are nevertheless constructed symmetrically over both hypotheses, to mimic the detector-level situation in which no such label exists. The per-particle tokens populate grades zero and one (App. D); we complement them with two *event-level pairing tokens*, one for each candidate b -assignment $a \in \{1, 2\}$, that inject grade-two and grade-three content of the multi-resonance topology directly at the input layer. The two tokens supplement the six per-particle tokens (eight tokens in total) and concretely realise the event-as-multivector view of Sec. 1 at the input rather than only through internal self-attention: their covariant channels carry the candidate hadronic-top trivectors $T_{\text{had}}^{(a)} = p_{j_1} \wedge p_{j_2} \wedge p_{b_a}$ (grade three), their Hodge duals $\star T_{\text{had}}^{(a)}$ (grade one), and the meet $T_{\text{had}}^{(a)} \cap T_{\text{lep}}^{(a)}$ of the two candidate top trivectors, with $T_{\text{lep}}^{(a)} = p_\mu \wedge p_{\bar{\nu}} \wedge p_{b_{\bar{a}}}$ built from the complementary b -jet, which is grade two because the dual of a grade-three blade is grade one and the wedge of two grade-one elements is grade two, so that the grade-two and grade-three covariants of both b -assignments enter the input simultaneously, the intention being that the equivariant attention head weights the two candidate assignments without an explicit combinatorial step. We report no assignment-accuracy measurement for that head, and Sec. 7.1 finds that these tokens do not improve the classifier. Their scalar channels (grade zero) carry the Breit–Wigner pulls (eq. (16)) of the two candidate top-mass reconstructions,

$$s_{\text{had}}^{(a)} \equiv \frac{(p_{j_1} + p_{j_2} + p_{b_a})^2 - m_t^2}{m_t \Gamma_t}, \quad s_{\text{lep}}^{(a)} \equiv \frac{(p_\mu + p_{\bar{\nu}} + p_{b_{\bar{a}}})^2 - m_t^2}{m_t \Gamma_t}, \quad \bar{a} \equiv 3 - a, \quad (16)$$

the symmetric and antisymmetric Wb combinations (writing $W_{\text{had}} \equiv p_{j_1} + p_{j_2}$ for the hadronic W candidate)

$$\sigma_{Wb}^{(\pm)} = \frac{1}{2}[(W_{\text{had}} + p_{b_1})^2 \pm (W_{\text{had}} + p_{b_2})^2],$$

and the analogous combinations for the leptonic W , the two candidate W -boson masses, the coplanarity scalar $\langle M\tilde{M} \rangle_0$ of the meet, and the pseudoscalar sign — ten channels in all (App. D). By Theorem 1 every scalar channel of the pairing tokens reduces to a polynomial in $\{p_i \cdot p_j, m_i^2\}$ and so does not constitute a new Lorentz invariant; the structural content of the pairing tokens is therefore the explicit appearance of the grade-two and grade-three covariant objects of the multi-resonance topology at the input layer, rather than their build-up inside the network.

What that explicit appearance buys is measured in Sec. 7.1: on this benchmark, whose entire class difference is one resonant three-particle combination, nothing — the collapse of Theorem 1 bounds what it could have been to zero new invariants.

Relation to prior work. At the per-particle level the token shape $T_i = \langle T_i \rangle_0 \oplus \langle T_i \rangle_1$ of GA coincides with the multivector input encoding of L-GATr [10]: grade-zero channels carry the object-type one-hot encoding and discrete tags, grade-one channels carry the four-momentum, and higher-grade covariants are built up inside the network by the geometric-product layers. Three properties distinguish the present construction from the geometric-algebra transformer line of work [8–10]. (i) *Event-level pairing tokens, and where they fail.* The two pairing tokens described above place grade-two and grade-three covariants of the multi-resonance topology at the input layer. The L-GATr line populates input bivector channels only with global-geometry reference tokens (beam axis, time direction) and reconstructs multi-particle topology content internally through the geometric-product layers; the present construction supplies that topology content directly at input instead. We had expected this to be an advantage. It is not:

the ablation of Sec. 7.1 shows that the topology tokens do not improve the classifier, while a fixed grade-two *reference* bivector — the beam plane, i.e. exactly the kind of object the L-GATr line already uses — improves it by 0.00751 ± 0.00055 , some 3.6 times what removing the topology tokens is worth, at identical parameter count. The distinction that survives is therefore not input-versus-internal but derived-versus-reference. One half of it is a theorem: a grade- k object reconstructible from the final-state momenta adds no invariant, by Theorem 1. The other half is a hypothesis, and we state it as one: content encoding structure absent from the measured momenta need not be redundant, and may be worth presenting in the grade that carries it. The evidence here is one instance of it — the beam bivector, in this final state with this architecture — which is what a first realisation on a minimal benchmark can supply; establishing the scope of the statement across processes and architectures is work for the programme, not for this example. Sec. 7.1 quantifies both halves of that statement. (ii) *No new invariants above grade one*. Theorem 1 bounds the scalar channel and identifies the grade-four pseudoscalar sign as the only genuinely new Lorentz-invariant content, separating representational from invariant additions across grades. (iii) *Grade-resolved fine-tuning interface*. The same per-token grade decomposition supplies a uniform read-out across classification, regression and CP-asymmetry tasks (Sec. 8). Fine-tuning in the L-GATr line proceeds by re-initialising the final output layer [10] without exposing per-grade outputs; the present interface instead routes each task to the grade that carries the relevant covariant content.

Reporting. The training is a binary classification of tW against $t\bar{t}$; we report a single ROC curve per network, with the area under the curve quoted next to each label. Every number is computed on events held back from training, in their natural rather than equalised class proportions; class imbalance is handled by a balanced sampler during training and enters the AUC only through its variance, not through the ranking statistic itself. We additionally report two discriminant-output histograms, one per network, each showing three curves obtained by passing through the trained network three independent parton-level samples: the single-resonant tW sample (Class A truth in the diagram-removal notation introduced above), the double-resonant $t\bar{t}$ sample (Class B), and the full gauge-invariant tWb matrix-element sample of ref. [11]. The full-schema sample is not used in training and is not treated as a separate class; it represents the realistic event population that combines both contributions and their interference, and the third curve illustrates how the network, trained on the cleanly-separated tW and $t\bar{t}$ distributions, behaves on this inclusive mixture as a function of the network output. The histograms are normalised by the matrix-element cross-sections of the three samples, so that the bin contents are $d\sigma/dD$ and the relative populations of the three curves reflect their physical weights rather than the Monte-Carlo statistics.

Prior expectations. By the Cayley–Menger collapse (Theorem 1), the scalar inner-product features of REF already span the full Lorentz-invariant scalar space accessible to any parton-level classifier on the inclusive sample; the asymptotic AUC of the two networks is therefore expected to be comparable. A gain of GA over REF, if any, is sought elsewhere: (i) the multi-resonance reconstruction problem of pairing the two b -jets to the candidate top resonances is exposed algebraically through the internal bivector and trivector channels and the event-level pairing tokens, so that the assignment is learned from a tighter representation rather than from scalar reconstruction shortcuts; and (ii) the algebraic representation supplies covariant access to all $p_i \wedge p_j$ and $p_i \wedge p_j \wedge p_k$ structures in a single forward pass, an inductive bias that is opaque in the scalar inner-product representation of REF.

AUC and ROC. The numerical outcome is summarised in Table 3 and Fig. 6. Both networks are trained and evaluated at parton level on the same generated samples ($\sim 270k$ tW and

Input representation	Params	AUC	s.e.	Δ AUC vs. four-momenta
grade 0 only	151 489	0.9627 ± 0.0003	0.0005	$-0.00422 \pm 0.00025^*$
+ pairing invariants	151 489	0.9650 ± 0.0003	0.0005	$-0.00195 \pm 0.00025^*$
+ pairing 1, 2, 3	151 489	0.9649 ± 0.0008	0.0005	$-0.00206 \pm 0.00031^*$
+ grade-4 join	196 609	0.9652 ± 0.0009	0.0005	$-0.00176 \pm 0.00060^*$
+ pairing 1, 3 (no meet)	151 489	0.9657 ± 0.0009	0.0005	$-0.00122 \pm 0.00039^*$
grade 0 $\oplus$ 1 four-momenta	149 409	0.9669 ± 0.0007	0.0005	—
+ ref. tokens γ_0, γ_3	149 409	0.9670 ± 0.0004	0.0005	$+0.00009 \pm 0.00021$
+ ref. bivector γ_{03}	149 409	0.9744 ± 0.0007	0.0004	$+0.00751 \pm 0.00055^*$
REF [11], 75 feat.	539 501	0.9586	0.0002	-0.0083

Table 3: Parton-level classification of single-resonant tW against double-resonant $t\bar{t}$ topologies in $pp \rightarrow tWb$ at $\sqrt{s} = 14$ TeV, $p_T^{j_4} > 10$ GeV. Rows differ only in what the input tensor carries. Each configuration is trained with five random seeds and scored on held-back events with final-epoch weights, so no model is selected by the quantity reported; column 5 differences each seed against its own four-momentum run, and an asterisk marks $|\Delta| > 2\sigma$. Parameter counts identify the comparable sets: a fixed reference token carries no weights, so it leaves the count unchanged, while the join channel stands alone. The REF row is a single training, so it carries no seed spread.

$\sim 10M$ $t\bar{t}$ events passing $p_T^{j_4} > 10$ GeV). For GA we report the mean and the spread over five independent training runs differing only in their random seed, which sets the weight initialisation, the train/validation split and the batch order together; configurations are compared seed by seed, so each paired difference is taken at a fixed split. Fig. 6 shows one of those runs, whose areas sit within that spread of the means in Tab. 3.

Discriminator distributions. At the discriminant level (Fig. 7) both input configurations resolve tW and $t\bar{t}$ cleanly, and the inclusive tWb -schema sample is bimodal under both, with mass between the two single-class peaks — the empirical signature of an event population that combines a single-resonant and a double-resonant component with their interference. The dominant peak at $D \rightarrow 1$ on the tW curve is a kinematic effect rather than a training artefact: it is populated almost entirely by events with $\sqrt{\hat{s}} < 2m_t$, where a $t\bar{t}$ pair is energetically inaccessible and the logit therefore saturates. It is present for both inputs and sharper with the reference bivector, which is the same effect the AUC of Tab. 3 records. The percent-level wrong-side tails are likewise a property of the truth definition rather than a training artefact: tW events in which both top-line invariants fall accidentally onto the Breit–Wigner pole ($m_t = 172.5 \pm 2$ GeV in both legs) are kinematically indistinguishable from $t\bar{t}$, and conversely $t\bar{t}$ events with one top-line strongly off-shell ($m_t \gtrsim 300$ GeV at the 95% percentile of the misclassified subsample) populate the same phase space as tW . The two populations interfere as gauge-invariant pieces of the same matrix element and the diagram-removal scheme leaves a small irreducible residue at the matrix-element level, consistent with the stability of the wrong-side fractions across independent training runs noted in the figure caption.

7.1 Which grades earn their place: an input ablation

Two questions are worth separating here, and the second is the narrower one. The first is whether a representation designed to be uniform across processes is competitive at all against features chosen for one process: the generic input of four-momenta and object-type labels, with no feature engineering for $pp \rightarrow tWb$, reaches AUC 0.9669 ± 0.0007 against 0.9586 for REF, whose input recipe was optimised for exactly this separation [4, 5, 11], at the parameter

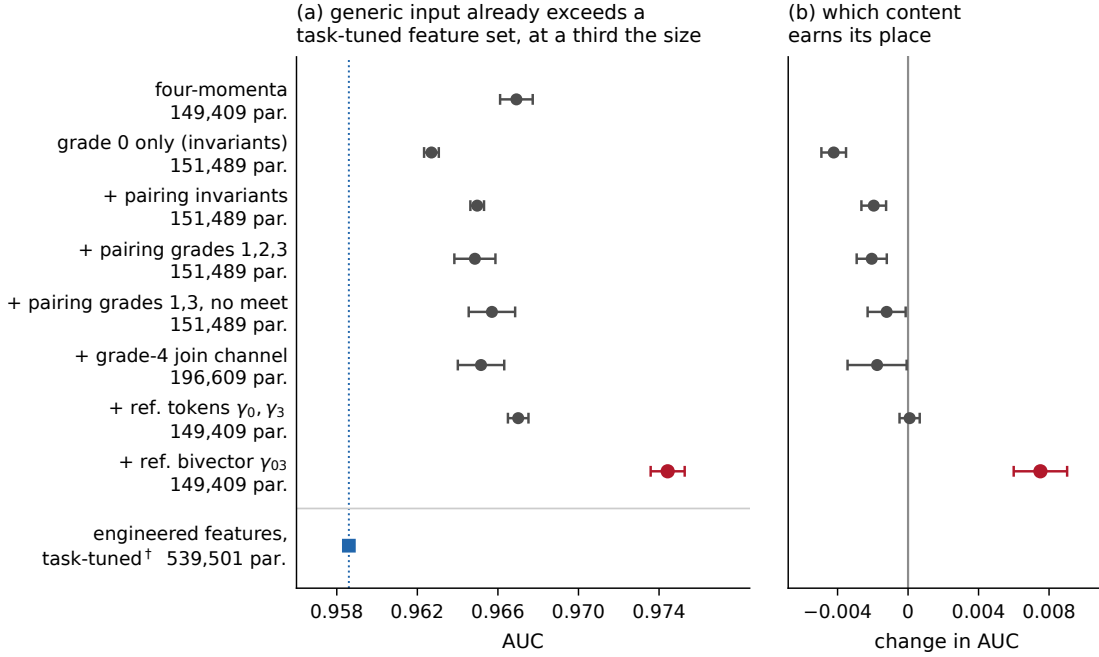


Figure 5: (a) Absolute performance of each input representation, with the engineered-feature baseline REF [4, 5, 11] for comparison; the generic algebraic input already exceeds it at a third the parameters (Sec. 7.1, Tab. 3). (b) The ablation proper: each representation against the four-momentum baseline, differences formed seed by seed and averaged, so that architecture, data and split are held fixed and only the input content varies. Trainable parameters are given beside each row; rows at equal count are internally comparable, and App. D gives the two input layouts cell by cell. Intervals are 95% Student intervals on four degrees of freedom.

ratio given above; with the fixed reference bivector the same network reaches 0.9744 ± 0.0007 . That is the comparison the universality claim has to survive, and it does.

The second question is which algebraic content earns its place once the representation is fixed. The construction of the preceding subsections puts grade-two and grade-three content at the input layer, and Sec. 1 motivates that choice representationally; whether it is worth anything empirically on *this* final state is answered by training the same network on a ladder of input representations differing in nothing else. The benchmark is the first and deliberately smallest realisation of the representation, so what follows measures the input choices here and does not settle them for other processes. Eight configurations are trained with five random seeds each; Tab. 3 collects the results and Fig. 5 displays them. The ladder has two branches, distinguished by their input layout rather than by nesting: the upper rows share the wide 22-channel projection of the pairing layout — the grade-0-only rung is that layout with every covariant cell zeroed, the ‘+’ rungs restore the four-momenta and vary the pairing content — while the lower rows use the 9-channel projection of the four-momentum baseline. The 2080-parameter difference between the groups (151 489 against 149 409) sits entirely in the width of that projection, while an added token costs no parameters at all; App. D lays out both formats cell by cell.

The effects under test are of the same order as the scatter between seeds, so comparing averages across seeds would be too noisy to read. Each pair of configurations is therefore differenced at a common seed, where both see identical events, and those differences are averaged. Every number quoted below is such a paired average.

Where the higher grades are formed. The contrast that isolates the question holds everything but the grade content fixed. The two pairing tokens stay, with their ten grade-zero invariants and the same 151 489 parameters; only their covariant channels — grades one, two and three — are zeroed. The difference is $+0.000115 \pm 0.000251$ in AUC, $t(4) = 0.46$, $p = 0.67$, with a 95% interval of $[-0.00058, +0.00081]$: nothing is resolvable at the precision five paired runs provide.

The reading this supports is about the *location* of the higher grades, not about their value. The network is Lorentz-equivariant on multivector channels, so when the grade-two and grade-three objects are not supplied it forms them from the momenta it does receive — attention mixes momenta across tokens, the geometric product wedges them within a token. The null result therefore says that on this final state the internal construction is as good as the explicit one, which is a statement that the network recovers the higher-grade content it needs, not that the content is dispensable. The direct evidence that higher-grade content is decisive appears in the same table: one fixed grade-two bivector, which no product of the measured momenta generates, is worth $+0.00751 \pm 0.00055$. A network indifferent to grade two could not respond to it. What the ablation locates is the boundary between the grade content a network can rebuild for itself and the grade content it cannot.

Two coarser removals do move the number. Dropping the pairing tokens outright gives $+0.00206 \pm 0.00031$ ($t(4) = 6.70$, $p = 0.003$); dropping only the grade-two meet while keeping the grade-three trivectors gives $+0.00085 \pm 0.00032$ ($t(4) = 2.62$, $p = 0.06$, interval $[-0.00005, +0.00174]$), a suggestive point estimate rather than an established effect. Neither isolates the grade content: removing the tokens also removes two sequence positions, narrows the input projection from 22 channels to 9, and changes both the attention graph and the mean pooling of the read-out. Whatever those two shifts measure, it is not the explicit higher-grade content on its own.

Theorem 1 accounts for the null result of the isolated comparison at the level of information: the covariant pairing channels carry no Lorentz-invariant quantity that is not already a function of the momenta the network receives. The theorem does not, however, state that removing redundant inputs is free for a network of finite capacity trained for a finite time — redundant channels can still help or hurt an optimisation — so it neither predicts the small shifts above nor is contradicted by them.

Why these grades, in this process. Which part of the multivector carries the discrimination is a property of the final state, not of the representation — and this benchmark was chosen so that the property is extreme: the class difference is carried mainly by one resonant three-particle combination. Here both W bosons are on shell in *both* classes, so the invariant mass of their grade-two decay planes is distributed alike: as single observables, $m(\mu^- \bar{\nu}_\mu)$ separates the classes with AUC 0.5077, the grade-two norm of the same plane gives the identical value (for massless partons it equals $(m^2/2)^2$), and the distance to the W pole gives 0.5004. These are one-dimensional tests of three particular functions of the plane; they leave open whether its orientation, or its correlation with the rest of the event, carries class information that a network could use. The leading class difference is a three-particle invariant, the leptonic top candidate, resonant only in the $t\bar{t}$ class, so grade three is the sector that matters here. Its bare mass, however, separates the classes only at 0.5189: a resonance sits inside the continuum it competes with, and the mass is not monotonic in “resonant or not”. As the pole distance $-|m - m_t|$ the same invariant gives 0.7442, and the two top candidates together give 0.9570. The grade- k object supplies the argument; the non-monotonic function of it is the network’s job. Grade four is populated, and its sign distribution in this sample is symmetric to within the sample precision (pseudoscalar AUC 0.5093); Sec. 8.1 explains why that is a check on the contraction order rather than a test of CP.

A different process moves the discriminating content to a different grade. A two-body resonance search — $H \rightarrow b\bar{b}$, $Z' \rightarrow \ell\ell$ — puts it in the grade-two plane of the pair, exactly the sector that is inert here. A CP-violating coupling, in $t\bar{t}$ production or in $H \rightarrow \tau\tau$, activates the grade-four sign, which no combination of inner products reproduces and which is the one channel Theorem 1 leaves open. Angular-correlation measurements of top-quark spin live in the grade-two bivector of the production plane. The ablation below should therefore be read as a measurement of which grades are informative *for this final state*; what generalises is the criterion, that a grade- k object reconstructible from the visible momenta adds no invariant, not the verdict on any particular grade.

A fixed reference bivector helps, and by much more. Adding a single fixed token carrying the beam bivector γ_{03} to the grade-0 $\oplus$ 1 configuration improves the AUC by $+0.00751 \pm 0.00055$ at identical parameter count. It is not the mere presence of a reference token that does this: the two *vector* reference tokens γ_0 and γ_3 give $+0.00009 \pm 0.00021$ over the same baseline, and the bivector-minus-vectors difference is $+0.00742 \pm 0.00047$, at equal parameter count. Equal capacity does not make the two configurations equal in every other respect — the bivector is one token where the vectors are two, so they differ in sequence length, in the attention graph and in the dilution of the mean-pooled read-out — so what the measurement establishes is that the H_{LHC} -invariant one-token bivector encoding outperforms the tested two-vector encoding in this architecture, rather than isolating the grade as the cause. Sec. 5 gives the reason we expected that ordering, and it is independent of the token count: γ_{03} is invariant under both generators of H_{LHC} to floating-point precision, whereas γ_0 and γ_3 separately are not, so the bivector realises the reduction by construction while the vector pair merely supplies its ingredients.

The controlling distinction is thus not the grade as such but its provenance. A grade- k object built from the measured momenta adds no Lorentz-invariant information, however explicitly it is presented — and on this benchmark its explicit presentation changed nothing measurable; a grade- k object encoding structure those momenta do not contain — here the laboratory frame singled out by the beam — does.

Reduced training statistics. An inductive bias is expected to pay off most where data are scarce, so the comparisons were repeated with the training set thinned to one third while the evaluation events were left identical. The epoch and warmup budgets were scaled with the thinning so that every fraction sees the same number of optimiser steps, which keeps the comparison about the data rather than the length of training. At 3.1M training events instead of 9.3M, and over three seeds, removing the candidate-pairing content gains $+0.00215 \pm 0.00069$ against $+0.00206 \pm 0.00031$ at full statistics, and the reference bivector gains $+0.00741 \pm 0.00058$ against $+0.00751 \pm 0.00055$. Each point estimate agrees with its full-statistics value to within a fifth of a standard error (0.12σ and 0.13σ). The reference-bivector gain is resolved at reduced statistics as well ($t(2) = 12.9$, $p = 0.006$); the pairing removal is not ($t(2) = 3.1$, $p = 0.09$, interval $[-0.00084, +0.00513]$), three runs being too few to separate it from zero. So there is no indication that either effect grows as data become scarce, but the reduced-statistics intervals are wide enough that a moderate growth of the pairing effect would not have been detected.

One further rung isolates the invariants. In the grade-zero-only limit the network sees Lorentz invariants alone; it loses 0.00422 ± 0.00025 against grade-0 $\oplus$ 1 four-momenta, so the covariant grade-one input is doing real work. The grade-four join row carries 32% more parameters (196 609 against 149 409) and still loses 0.00176 ± 0.00060 against the same baseline, so on this final state the join channel does not pay for its extra capacity; it is also the one row not capacity-matched to any other.

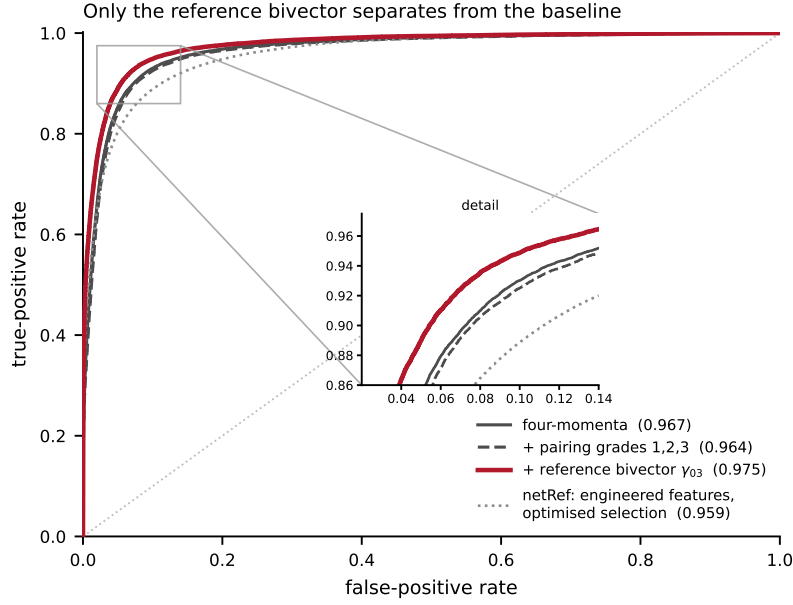


Figure 6: Receiver-operating curves on the parton-level discrimination of single-resonant tW against double-resonant $t\bar{t}$ topologies of $pp \rightarrow tWb$ with $p_T^{j_4} > 10$ GeV, for three input representations of the same network. The inset magnifies the region where they separate: adding the candidate-pairing content built from the final-state momenta leaves the curve on top of the four-momentum baseline, while one fixed reference bivector γ_{03} lifts it. Curves are one seed of five. The lower grey curve is REF [11], the engineered-feature baseline network on the hand-optimised input recipes of refs. [4, 5]. The dotted diagonal is the random classifier.

Code and reproducibility. The training code for GA used in this section, together with the final version of this paper and instructions for obtaining the Monte-Carlo samples, is available in the public companion repository [43]. The repository collects the reference implementations of the algebraic input layer and of the networks built on it across the foundation-model programme of which the present demonstration is the first instalment.

8 Foundation models for collider events

Foundation models in collider physics are now starting to be explored [19]; the present algebraic representation is intended as their input layer. We close the paper by sketching several directions in which the multivector $\mathcal{E}$ provides a uniform input layer for the foundation-model program of which the present specification is the first piece.

What the algebraic representation buys. Three properties of $\mathcal{E}$ make it a natural foundation-model input layer. (i) *Task-independence.* The multivector $\mathcal{E}$ of eq. (4) is generic over the choice of physics process: the same per-event object encodes W^+W^- , $t\bar{t}$, $H \rightarrow b\bar{b}$, EFT signals and BSM search regions, with only the basis decomposition on V_{flav} varying. New states enter through V_{flav} , which acquires basis elements for them, while modified interactions of Standard-Model particles — the SMEFT case, in which high-scale physics appears as shifted couplings and new Lorentz structures rather than as new particles — enter through the kinematics, i.e. through the grade content of $\mathcal{E}$ at fixed V_{flav} . Both routes stay inside the same four-dimensional

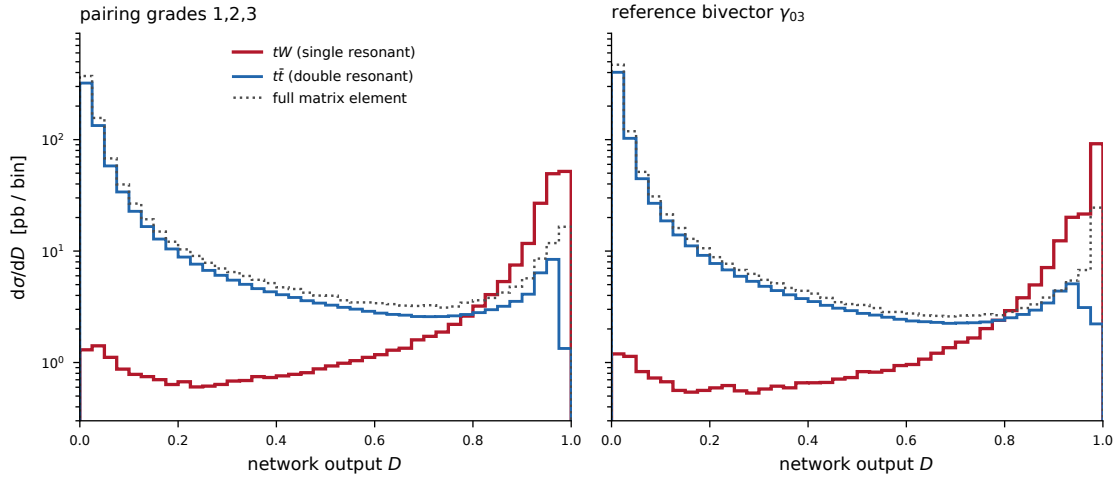


Figure 7: Discriminator distributions, normalised to the parton-level cross sections, for two input representations of the same network: the candidate-pairing content of grades one to three (left) and the fixed reference bivector γ_{03} (right). The single-resonant tW and double-resonant $t\bar{t}$ samples are separated cleanly by both, and the full matrix element tracks the double-resonant shape, as the interference region should. The bivector sharpens the tW peak: 58% of tW events have $D > 0.95$ against 52% on the left, and 68% of $t\bar{t}$ events have $D < 0.05$ against 62%.

spacetime algebra, which is why one representation serves both. The grade decomposition supplies a tokenisation that survives across tasks, so a pre-trained encoder is fine-tuned by attaching a task head. (ii) *Lorentz covariance throughout*. Pre-training on $\mathcal{E}$ preserves grade structure end-to-end: a downstream task requiring strict invariance reads grade-zero coefficients of the last hidden state, while a task requiring full covariance reads the full multivector. This contrasts with pre-training on lab-frame recipes [4, 5], where the frame choice propagates to every downstream task, and with pre-training on raw four-momenta without an algebraic structure, where every task has to relearn the representation of $\text{Spin}^+(1, 3)$. (iii) *H_{LHC} -equivariant sub-decomposition*. The symmetry actually available at a hadron collider is H_{LHC} (eq. (11)), because the beam axis singles out a direction; missing transverse momentum adds a second restriction, the unmeasured longitudinal component of the neutrino system. Both are encoded as a sub-decomposition of $\mathcal{E}$ in which $\text{SO}^+(1, 1)_z$ acts on the grade-one (y, M) sector and $\text{SO}(2)_\phi$ on the grade-one (p_T, ϕ) sector; the missing momentum entering as a pseudo-particle of partially-populated grade one with its p_z flagged in V_{flav} . Of the three mechanisms that realise the reduction — the fixed γ_{03} reference token (whose invariance is eq. (13)), a layer structure adapted to the 2+2 split of eq. (12), and the masked grade-one pseudo-particle — only the first is implemented and measured in this paper (Sec. 7.1); the other two are proposals. The second is the least developed of the three: block-diagonal weights respecting the 2+2 split are a necessary condition but not a sufficient one, since a layer equivariant under H_{LHC} requires each grade to be decomposed into H_{LHC} -irreducible components and the weights to act as intertwiners between them. Working that decomposition out for all five grades is a construction we have not carried out.

Relation to existing work. Existing collider foundation-model proposals — masked-particle pre-training [44], contrastive jet self-supervision [45] and multi-task event models [19] — operate on raw four-momenta plus flat auxiliary scalar channels for particle type, charge, b -tag and process label, which are exactly the content of V_{flav} here; embedding them in $\text{Cl}(1, 3) \otimes V_{\text{flav}}$

makes them one factor of a single structure instead of case-specific additions, with the same token interface in pre-training and in every downstream task. Those proposals also impose no architectural equivariance, whereas the backbone assumed here is the multivector-equivariant transformer of refs. [9, 10] on the same algebra, which adopts multivector inputs but trains single-task, naming cross-task pre-training as future work; the invariant-aggregator PELICAN line [7, 20] spans the grade-zero sector $\langle \mathcal{E} \rangle_0$ alone, and Sec. 7.1 measures what the covariant grades add over it. The explicit grade factorisation is what makes the masking targets and per-grade heads above available at all.

Grade-resolved pre-training and read-out. The interface the grade decomposition supplies is a masking target and a read-out head per grade. Pre-training predicts masked components of $\mathcal{E}$ from the unmasked ones, and each grade exposes different physics: grade zero the Lorentz invariants, grade one the mass-shell constraints and matrix-element correlations, grade two the decay-plane geometry, grade three the T -odd oriented volume [18, 35], grade four the CP structure — so the per-target loss curves report which content the encoder has internalised. Masked pairs involving a neutrino use the truth p_ν at parton level and the E_T^{miss} pseudo-particle of Sec. 6 at reconstruction level. Fine-tuning then attaches a single head to the unchanged encoder: classification and scalar regression read the grade-zero coefficients of the last hidden state, four-momentum regression the grade-one component, and a CP-asymmetry test the grade-four coefficient, whose sign expectation is compared against a CP-even null.

8.1 Practical obstacles

The preceding sketch is an argument that the representation is a suitable input layer, not that building on it is straightforward. Four obstacles are specific to a graded representation, and we state them with numbers from our own runs rather than as generic caveats, because in each case the measurement changes what the obstacle is.

Per-grade dynamic range. An object built by wedging k momenta carries mass dimension k , so the numerical scale of the input channels is a property of the algebra rather than a nuisance to be normalised away. In the pairing tokens of Sec. 7 the raw entries run from $\mathcal{O}(10)$ GeV² at the low tail of a grade-two invariant to $\sim 7 \times 10^{25}$ GeV¹² at the high tail of the coplanarity scalar — the squared norm $\langle M\tilde{M} \rangle_0$ (eq. (6)) of the meet $M = T_{\text{had}} \cap T_{\text{lep}}$, six powers of momentum in each factor — i.e. 25 orders of magnitude inside one input tensor. That the spread is dimensional and not an artefact is checkable on typical rather than extreme values: each object’s median taken to the power $1/k$ returns 43–190 GeV throughout, a parton-scale energy at every mass dimension from one to twelve. Per-grade normalisation removes the spread between objects, but within one object up to 6 orders of magnitude remain, so a network must either learn across that range or be given a grade-aware scaling — and a foundation model shared across processes cannot fix these scales at build time, since they follow the mass and $\hat{s}$ scales of whatever final state it meets.

The pseudoscalar sign must survive normalisation. The one Lorentz-invariant channel not reducible to inner products is the *sign* of the grade-four pseudoscalar (Theorem 1). Any normalisation acting on magnitudes — per-grade rescaling, layer normalisation, a log compression of the range just described — risks discarding it, and a one-bit observable is not recoverable once lost. This constrains the normalisation scheme rather than being a matter of taste: the transformation applied to the grade-four channel must be sign-preserving.

Flavour labelling can manufacture a false CP asymmetry. The pseudoscalar sign is advertised as a CP-odd observable, which makes it a trap. The symmetry test available to us is a test of the contraction order, not of CP: our sample is the fixed μ^- channel, and CP maps that channel onto its charge-conjugate μ^+ counterpart, which we do not generate — so a symmetric sign distribution here is not a consequence of CP conservation, and an asymmetric one would not be evidence against it. What it does diagnose is whether the ordering convention imprints itself on the sign. With the two same-parent light partons ordered by energy the positive fraction is 0.4989 on the 268k-event tW sample, 1.2σ from one half. Contracting instead the PDG-labelled slots $\epsilon(p_\mu, p_{\bar{\nu}}, p_u, p_{\bar{d}})$ gives 0.5115, 11.9σ from one half — an apparent asymmetry of high significance produced by the labelling alone, of exactly the size and shape a CP-violation search would report as a signal. The mechanism is that exchanging the two partons of the same parent flips the sign of the contraction exactly, so a labelling that is correlated with the kinematics imprints itself on the sign. Any analysis using this channel must fix the contraction order by a physical rule and verify the symmetry on a charge-conjugate pair of samples, which is the comparison that does test CP; a truth-flavour ordering will not do, and a single-charge sample cannot settle the question either way.

Cost, and what it buys. Attention over multivector channels costs what the extra channels cost, and the count grows combinatorially with the final-state multiplicity: the candidate objects number $\binom{N}{2}$ bivectors, $\binom{N}{3}$ trivectors and $\binom{N}{4}$ pseudoscalars, i.e. 50 for the $N = 6$ partons of the present tWb final state, 154 for a $2 \rightarrow 8$ process, and 781 for the $N = 12$ partons of four-top production or $t\bar{t}tWb$ — fifteen times the present case for a process now under active study. In our runs the two pairing tokens alone raise the epoch time from 259 to 361 s against the four-momentum baseline of Sec. 7.1, that is by 39%, and that section shows they also lower the AUC. The reference bivector that does help adds no parameters and no measurable time, since it is one fixed token independent of N . The practical lesson is not that graded inputs are expensive but that their cost grows with the combinatorics while their benefit is task-dependent: on a single task an implementation should spend its budget on the sectors that task reads — here, reference content the momenta do not contain — whereas a foundation model spanning processes cannot prune a priori and must budget for the full decomposition, deferring the pruning to fine-tuning.

9 Limitations and future directions

What the construction does and does not do. The construction is bound by the Cayley–Menger collapse (Theorem 1; grade-four paragraph of Sec. 3); pre-Lorentz-frame observables such as sphericity, thrust and aplanarity sit in an orthogonal complement to the bivector channel (event-shape caveat in Sec. 3); b -tag information is reduced to the binary slot $|bj\rangle \in \mathbb{F}$ (Sec. 6); fermion spin states enter implicitly through the bivector grade and the second-moment structure of production and decay [32, 34, 35], with no explicit spinor-valued fields. These are scope choices, not deficiencies of the algebra.

Conformal $\text{Cl}(2,4)$ for multi-resonance event topologies. A physical motivation for an algebra beyond $\text{Cl}(1,3)$ appears as soon as one considers events with more than one heavy intermediate resonance: a $t\bar{t}$ event, a four-top event, or a ladder of W -mediated cascades. In each such case the kinematic support of the event is naturally described by the *intersections* of mass shells $(p_a + p_b)^2 = M^2$ in four-momentum space, but in $\text{Cl}(1,3)$ those intersections are forced into scalar Cayley–Menger combinations of inner products (Theorem 1); the geometric content of “one resonance is hit, the other is missed” is dissolved into scalars even

though physically the two configurations populate distinct sub-manifolds of phase space. The conformal embedding into $\text{Cl}(2, 4)$ [1] keeps this content algebraic. Two auxiliary null vectors n_∞, n_o extend the generating space to six dimensions, and each four-momentum is mapped to the null conformal vector

$$P = n_o + p + \frac{1}{2} p^2 n_\infty, \quad P_i \cdot P_j = -\frac{1}{2} (p_i - p_j)^2, \quad (17)$$

so that the Lorentz-invariant squared four-momentum difference $(p_i - p_j)^2$ between any two particles is the inner product of their conformal images [1, §10.2]. A mass shell $(p_W + p_b)^2 = m_t^2$ then becomes one blade of fixed grade [1, 2] — a geometric primitive rather than a side equation. The meet of two such shells, $\Sigma_{t_1} \cap \Sigma_{t_2} = (\Sigma_{t_1}^* \wedge \Sigma_{t_2}^*)^{-*}$, is a lower-grade blade in the conformal algebra carrying the kinematic intersection of the two top hyperboloids; its squared norm has a definite sign that distinguishes events in which two real on-shell top assignments are kinematically possible from configurations at threshold and from configurations with no real intersection. The same construction generalises to any chain of nested heavy resonances and is the natural multi-resonance generalisation of the single-bivector decay plane of Sec. 3. In the same conformal language, the combinatorial b -assignment of Sec. 7 reduces to a single incidence test $(W + p_{b_a}) \cdot \Sigma_t \approx 0$ between the conformal image of the $(W + b_a)$ candidate sum and the top mass-shell blade Σ_t ; the test takes the same form for arbitrary N -resonance topologies. This is a representational restatement of the multi-resonance reconstruction problem that motivates the event-level pairing tokens of Sec. 7. On the tWb task it is not an empirically useful one (Sec. 7.1), the intersection observables being functions of the Gram matrix. It may still pay off where the incidence test is *not* reconstructible from the visible momenta — topologies with more than one unmeasured momentum — and that is the direction we would look next. By contrast, the interface content the same measurement does support is the fixed reference object: a foundation model built on this algebra should carry the beam plane γ_{03} as a fixed reference blade: it is the (t, z) plane, rather than the time and beam directions separately, that is H_{LHC} -invariant (eq. (13)) and that the measurement of Sec. 7.1 favours. What the conformal embedding does *not* do is enlarge the ring of Lorentz-invariant scalars: by the First Fundamental Theorem of invariant theory for the orthogonal group [33], every Lorentz-invariant scalar built from the conformal vectors $\{P_i\}$ together with the fixed null pair (n_o, n_∞) reduces to a polynomial in the inter-particle products $\{P_i \cdot P_j\} = \{-\frac{1}{2}(p_i - p_j)^2\}$ and the mass projections $\{P_i \cdot n_o\} = \{-\frac{1}{2}m_i^2\}$ (with $P_i \cdot n_\infty = -1$ trivially), which together are polynomial in $\{p_i \cdot p_j, m_i^2\}$, and to the orientation sign of the grade-four pseudoscalar of Theorem 1. The conformal extension is therefore not a source of new Lorentz invariants beyond those already covered by Theorem 1; it is a *representational* extension, in which the multi-resonance geometry of the event is exposed as native algebraic objects whose use as input features to an equivariant network is the natural generalisation of the bivector attention layer of Sec. 7 to nested resonance topologies.

Other algebraic extensions deferred. The multi-particle Clifford bundle $\text{Cl}(1, 3)^{\otimes N}$ (Variant C of Sec. 6) makes the per-token factorisation explicit at the price of an N -dependent algebra (Variant B was analysed and rejected in Sec. 6 and is not deferred). The self-duality decomposition of grade-two multivectors and the Weyl-spinor parallel via the isomorphism $\text{Spin}^+(1, 3) \simeq \text{SL}(2, \mathbb{C})$, the hyperbolic geometry of the rapidity-azimuth plane $\mathbb{H}^1 \times \mathbb{S}^1$, and the projective extension $\text{Cl}(1, 3, 1)$ for an explicit vertex-level encoding are open directions to be evaluated in dedicated downstream studies.

Demonstration caveats. The $pp \rightarrow tWb$ illustration of Sec. 7 is a parton-level leading-order matrix element without parton shower or detector simulation; it is intended as a methodological demonstration of the algebraic input representation, not as a phenomenological es-

timite. The CP-odd channel identified in Sec. 3 is a one-bit observable whose expectation vanishes in CP-symmetric tree-level configurations, and a quantitative measurement at the network level requires CP-asymmetric matrix-element samples not used in the present demonstration. Cross-process generalisation, larger and more diverse training samples, fast- or full-simulation extensions, and a systematic comparison against representative Lorentz-equivariant baselines [8–10] are independent directions whose evaluation requires dedicated downstream tasks.

Towards the HiGEN foundation-model program. The construction provides the input-layer specification for the HiGEN program of Sec. 8, in which an event multivector $\mathcal{E}$ is the per-token input for a Lorentz-equivariant collider foundation model pre-trained across processes. The architectural details of the encoder, the per-grade pre-training strategy, the catalogue of downstream tasks (classification, regression, generative reconstruction) and the cross-process transfer-learning protocol are the subject of separate forthcoming work; the present paper fixes the algebraic framework on which that program is built.

10 Summary

A unified algebraic representation $\mathcal{E}$ (eq. (4)) has been introduced as a common framework for three previously disjoint feature-engineering approaches [4, 5, 10]. The physical meaning of every grade and operation has been catalogued, an explicit per-grade dictionary of 32 classical observables is supplied (App. C.1), and the spacetime, discrete and approximate symmetries acting on $\mathcal{E}$ are listed (App. C.2). Theorem 1 settles the question of new Lorentz-invariant scalars: none are unlocked beyond $\{p_i \cdot p_j, m_i^2\}$, and the genuine non-trivial channel is the CP-odd sign of the pseudoscalar [18]. Several directions in which $\mathcal{E}$ could serve as the input-layer specification for a foundation-model programme have been sketched (Sec. 8); their concrete elaboration is left to dedicated follow-up work. Four obstacles specific to a graded input are quantified on our own runs (Sec. 8.1): the 25 orders of magnitude of per-grade dynamic range, the sign of the grade-four channel that any magnitude normalisation can destroy, the false CP asymmetry that a label-ordered contraction manufactures, and the combinatorial cost of materialising the higher grades. The methodology has been illustrated on the resonance-topology separation of $pp \rightarrow tWb$ (Sec. 7; see also Sec. 9 for limitations).

Acknowledgements

Funding information This study was conducted within the scientific program of the National Center for Physics and Mathematics, section #5 “Particle Physics and Cosmology”. Stage 2026–2027.

A Notation card

For convenience we collect here the conventions used throughout the paper.

- **Metric signature.** $\eta = \text{diag}(+, -, -, -)$.
- **Basis of $\text{Cl}(1, 3)$.** Orthonormal grade-one basis $\{\gamma_0, \gamma_1, \gamma_2, \gamma_3\}$ with $\gamma_\mu \gamma_\nu + \gamma_\nu \gamma_\mu = 2\eta_{\mu\nu}$.
- **Pseudoscalar.** $I \equiv \gamma_0 \gamma_1 \gamma_2 \gamma_3$, $I^2 = -1$.

- **Reverse.** $\tilde{X}$ reverses the order of all geometric products inside X ; on an r -blade it acts as multiplication by the sign $(-1)^{r(r-1)/2}$.
- **Dual.** $X^* \equiv X I^{-1}$. The inverse map satisfies $X^{-*} \equiv X I$, so that $(X^*)^{-*} = X$; both are used in eq. (9). Our convention $X^* \equiv X I^{-1}$ differs from $X^* = X I$ used in ref. [1] by an overall sign, since $I^{-1} = -I$ in our signature.
- **Grade projection.** $\langle X \rangle_k$ is the grade- k component of X .
- **Geometric product.** XY . Inner product $X \cdot Y = (XY + YX)/2$ for grade-one elements; outer product $X \wedge Y = (XY - YX)/2$ for grade-one elements. The grade-one decomposition reads

$$pq = (p \cdot q) + (p \wedge q), \quad p \cdot q = \frac{1}{2}(pq + qp), \quad p \wedge q = \frac{1}{2}(pq - qp). \quad (\text{A.1})$$

In component form, for $p = p^\mu \gamma_\mu$ and $q = q^\nu \gamma_\nu$, symmetrising and antisymmetrising the index pair (μ, ν) in $pq = p^\mu q^\nu \gamma_\mu \gamma_\nu$ gives

$$pq = \eta_{\mu\nu} p^\mu q^\nu + \sum_{\mu < \nu} (p^\mu q^\nu - p^\nu q^\mu) \gamma_\mu \gamma_\nu, \quad (\text{A.2})$$

where the scalar (grade-zero) part is the Minkowski inner product $p \cdot q = p^0 q^0 - \vec{p} \cdot \vec{q}$ and the bivector (grade-two) part splits into three boost components $(p^0 q^i - p^i q^0) \gamma_0 \gamma_i$ ($i = 1, 2, 3$) and three rotation components $(p^i q^j - p^j q^i) \gamma_i \gamma_j$ ($1 \leq i < j \leq 3$), exhausting the six oriented planes of $\langle \text{Cl}(1, 3) \rangle_2$.

- **Blades.** An r -blade is the outer product of r linearly independent grade-one elements; in $\text{Cl}(1, 3)$ non-simple multivectors of grade r exist only for $r = 2$ (grade-three and grade-four elements are always blades by Hodge duality with grades one and zero, respectively).
- **Plücker line coordinates.** The raw covariant coefficients of a bivector $p_i \wedge p_j$ coincide, up to basis choice, with the Plücker line coordinates of the corresponding decay plane in projective $\mathbb{RP}^3$ [1].
- **Lorentz group and rotor sandwich.** Proper orthochronous Lorentz group $\text{SO}^+(1, 3)$, with double cover $\text{Spin}^+(1, 3) \simeq \text{SL}(2, \mathbb{C})$. A Lorentz transformation acts on *any* multivector, of any grade, by the same two-sided product

$$X \longmapsto R X \tilde{R}, \quad R = \exp(B/2), \quad B \in \langle \text{Cl}(1, 3) \rangle_2, \quad R \tilde{R} = 1,$$

where $\tilde{\cdot}$ is the reverse of App. B. One expression thus replaces the separate 4×4 , rank-two and rank-three transformation laws of tensor notation: the boost or rotation is specified once, by the bivector B in its plane, and applied unchanged to momenta (grade one), decay planes (grade two) and oriented volumes (grade three). Two uses recur in the text. First, changing frame: with R the rotor of the boost into a top rest frame, $\cos \theta^*$ is the grade-zero part of a product of rotated momenta (Sec. 3), and no separate frame-specific formula is needed per observable. Second, testing invariance: a candidate input is invariant under a subgroup precisely when it commutes with the corresponding rotors, which is how the reference bivector γ_{03} is shown to be H_{LHC} -invariant (eq. (13)) while γ_0 and γ_3 separately are not. Composition is rotor multiplication, $R_2 R_1$, and the inverse is $\tilde{R}$; the grade of X is preserved because the sandwich is an algebra automorphism (App. B).

- **Bivector sign rules.** For $B \in \langle \text{Cl}(1,3) \rangle_2$ the sign $\text{sgn}(B^2)$ separates rotation generators ($B^2 < 0$ in our signature, e.g. $(\gamma_1\gamma_2)^2 = -1$) from boost generators ($B^2 > 0$, e.g. $(\gamma_0\gamma_3)^2 = +1$); a generic bivector is the sum of one rotation and one boost in two mutually orthogonal commuting planes [1].
- **LHC residual subgroup.** $H_{\text{LHC}} \equiv \text{SO}^+(1,1)_z \times \text{SO}(2)_\phi \subset \text{SO}^+(1,3)$.
- **Event multivector.** $\mathcal{E} \in [\text{Cl}(1,3) \otimes V_{\text{flav}}]^{\oplus N}$ defined in eq. (4): one slot per reconstructed object.
- **Object-type space.** $V_{\text{flav}} = \text{span}_{\mathbb{R}}\{|f\rangle : f \in \mathbb{F}\}$ with $\mathbb{F}$ given by eq. (14).
- **Missing transverse momentum.** E_T^{miss} ; pseudo-object encoded by eq. (15).

B Algebraic derivations

Sign of $(p \wedge q) \widetilde{(p \wedge q)}$ in $(+, -, -, -)$. For two grade-one elements $p, q \in \text{Cl}(1,3)$ the bivector $B \equiv p \wedge q$ satisfies $\widetilde{B} = -B$ since reverse acts as $(-1)^{r(r-1)/2}$ on a grade- r element, and a direct expansion in $(+, -, -, -)$ gives

$$(p \wedge q)^2 = (p \cdot q)^2 - p^2 q^2, \quad (p \wedge q) \widetilde{(p \wedge q)} = -(p \wedge q)^2 = p^2 q^2 - (p \cdot q)^2, \quad (\text{B.1})$$

the second equality following from $\widetilde{(p \wedge q)} = -(p \wedge q)$ for any grade-two element [1]. For two future-directed time-like p, q the reverse Cauchy–Schwarz inequality $(p \cdot q)^2 \geq p^2 q^2$ holds, so $B\widetilde{B} \leq 0$ in our signature, and the strictly non-negative Lorentz scalar associated with the two-blade is

$$|p \wedge q|^2 \equiv -(p \wedge q) \widetilde{(p \wedge q)} = (p \cdot q)^2 - p^2 q^2 \geq 0. \quad (\text{B.2})$$

A consistency check of the identity (B.1) on the orthonormal basis pair $p = \gamma_0$ (time-like) and $q = \gamma_1$ (space-like) gives $B = \gamma_0\gamma_1$, $B^2 = +1$, $B\widetilde{B} = -1$, and $p^2 q^2 - (p \cdot q)^2 = (1)(-1) - 0 = -1$; the two sides agree, but this mixed-signature configuration lies outside the time-like time-like case in which non-negativity of $|B|^2$ was established. The time-like time-like case itself follows by continuity in the parameterisation $q = \cosh \alpha \gamma_0 + \sinh \alpha \gamma_1$, for which

$$p^2 q^2 - (p \cdot q)^2 = 1 - \cosh^2 \alpha = -\sinh^2 \alpha, \quad |B|^2 = \sinh^2 \alpha \geq 0.$$

Gram-determinant identity for an r -blade. For a generic r -blade $P_{i_1 \dots i_r} \equiv p_{i_1} \wedge \dots \wedge p_{i_r}$ constructed from r four-momenta the squared magnitude collapses onto an $r \times r$ Gram determinant of the participating inner products,

$$P_{i_1 \dots i_r} \widetilde{P}_{i_1 \dots i_r} = \det[p_{i_a} \cdot p_{i_b}]_{a,b=1}^r, \quad (\text{B.3})$$

with the right-hand side carrying its own sign through the Minkowski inner products and reproducing eq. (B.1) for $r = 2$. To derive eq. (B.3) choose an orthonormal frame $\{e_a\}_{a=1}^r$ of the subspace spanned by $\{p_{i_a}\}$, with $e_a \cdot e_b = \varsigma_a \delta_{ab}$ and $\varsigma_a = \pm 1$ encoding the signature of the subspace inside Minkowski space. Write $p_{i_a} = M^a_b e_b$ for a non-singular real matrix M . Then $P_{i_1 \dots i_r} = (\det M) e_1 \wedge \dots \wedge e_r$, and

$$P_{i_1 \dots i_r} \widetilde{P}_{i_1 \dots i_r} = (\det M)^2 (e_1 \wedge \dots \wedge e_r)(e_r \wedge \dots \wedge e_1) = (\det M)^2 \prod_{a=1}^r \varsigma_a.$$

The Gram matrix of $\{p_{i_a}\}$ reads $[p_{i_a} \cdot p_{i_b}] = M \operatorname{diag}(\varsigma) M^\top$, with determinant $(\det M)^2 \prod_a \varsigma_a$, which reproduces eq. (B.3) without any extra signature factors: the sign of the result is the sign of the Gram determinant in our signature.

The collapse result is stated as Theorem 1 in Sec. 3; its proof follows.

Proof. Single-grade contractions $\langle X_r \widetilde{X}_r \rangle_0$ reduce directly to inner products. For $r = 2$, eq. (B.1) gives

$$(p \wedge q) \widetilde{(p \wedge q)} = p^2 q^2 - (p \cdot q)^2,$$

and for $r = 3, 4$ the Gram-determinant identity (eq. (B.3)) yields, for an r -blade $P = p_1 \wedge \cdots \wedge p_r$,

$$P \widetilde{P} = \det[p_a \cdot p_b]_{a,b=1,\dots,r},$$

which is a polynomial in the inner products. A general grade- r multivector is a sum of blades, and any Lorentz scalar built from such a multivector is a polynomial in the inner products by linearity. The pseudoscalar $\langle P_4 \rangle_4 = c_4 I$ contributes only via c_4^2 to single-grade scalar combinations, and $c_4^2 = -\det[p_a \cdot p_b]_{a,b=1,\dots,4}$ by the Gram identity, which reduces to inner products as well.

For mixed-grade contractions $\langle X_{r_1} \cdots X_{r_n} \rangle_0$ the geometric product of two homogeneous-grade multivectors expands grade-by-grade as [1, eq. (4.42)]

$$X_r X_s = \sum_{\substack{k=|r-s| \\ k \equiv r+s \pmod{2}}}^{r+s} \langle X_r X_s \rangle_k.$$

For a chain of length two the grade-zero projection $\langle X_{r_1} X_{r_2} \rangle_0$ vanishes whenever $r_1 \neq r_2$, because the geometric product of two homogeneous-grade multivectors of distinct grades has no grade-zero component (the grades on the right run in steps of two from $|r_1 - r_2|$ to $r_1 + r_2$, and zero is hit only when $r_1 = r_2$). The surviving case $r_1 = r_2 = r$ defines the bilinear pairing $(X_r, Y_r) \mapsto \langle X_r \widetilde{Y}_r \rangle_0$ on grade- r multivectors [1, §4.1.3], related to $\langle X_r Y_r \rangle_0$ by the reversion sign $\langle X_r Y_r \rangle_0 = (-1)^{r(r-1)/2} \langle X_r \widetilde{Y}_r \rangle_0$. The polarisation identity

$$2\langle \widetilde{A} B \rangle_0 = \langle (A+B) \widetilde{(A+B)} \rangle_0 - \langle A \widetilde{A} \rangle_0 - \langle B \widetilde{B} \rangle_0,$$

applied with A, B both of grade r , then expresses $\langle X_{r_1} \widetilde{X}_{r_2} \rangle_0$ as a difference of three single-grade self-contractions $\langle Z \widetilde{Z} \rangle_0$ of grade r . For longer chains, induction on n closes the reduction, over the hypothesis that every grade- r projection $\langle X_{r_2} \cdots X_{r_n} \rangle_r$ of a shorter chain is a combination of r -blades of the p_i and their I -duals with coefficients polynomial in the inner products and at most linear in the λ_T (the base case $n = 1$ is a blade with unit coefficient). The duals must be admitted because a product can leave the blade class through the pseudoscalar: for instance $\langle (p_1 \wedge p_2 \wedge p_3 \wedge p_4)(p_5 \wedge p_6 \wedge p_7) \rangle_1 = \lambda_{1234} (I (p_5 \wedge p_6 \wedge p_7))$, a dual vector that is not an outer product of the momenta. The class is closed: I commutes through even elements and anticommutes with odd ones, so a product of two class members normal-orders to at most one explicit I factor, $I^2 = -1$ removes pairs, and a leftover single I landing in a grade-zero projection contributes $\langle I Y_4 \rangle_0 = -\lambda_Y$, which keeps the λ -linear form. Writing $\langle X_{r_1} \cdots X_{r_n} \rangle_0 = \langle X_{r_1} (X_{r_2} \cdots X_{r_n}) \rangle_0$ and expanding the right factor grade-by-grade, the chain-of-two argument selects the single grade $t = r_1$, so that $\langle X_{r_1} \cdots X_{r_n} \rangle_0 = \langle X_{r_1} \langle X_{r_2} \cdots X_{r_n} \rangle_{r_1} \rangle_0$ reduces to the bilinear pairing of two grade- r_1 objects, and the inductive hypothesis applied to $\langle X_{r_2} \cdots X_{r_n} \rangle_{r_1}$ (a grade- r_1 multivector built from a product of length $n - 1$) expresses every contribution as a polynomial in single-grade contractions $\langle Z \widetilde{Z} \rangle_0$. The grade-zero projection of any product of homogeneous-grade multivectors is therefore a polynomial in single-grade scalar contractions $\langle Y_t \widetilde{Y}_t \rangle_0$, which reduce to inner products by the previous paragraph, and in pseudoscalar coefficients λ_T arising whenever a grade-four projection appears in the reduction — whether from a grade-four factor X_a

or from the grade-four part of a product of lower-grade factors, as in $\langle(p_1 \wedge p_2)(p_3 \wedge p_4)\rangle_4$. The relevant parity is thus the number of λ factors after the reduction, not the number of grade-four X_a . Every even power and every pairwise product collapses by the *mixed* Gram identity

$$\langle(p_1 \wedge \cdots \wedge p_r)(q_1 \wedge \cdots \wedge q_r)\rangle_0 = \det(p_a \cdot q_b)_{a,b=1,\dots,r}, \quad (\text{B.4})$$

of which eq. (B.3) is the diagonal case $q_a = p_a$ (both sides are multilinear and antisymmetric in each argument set, so they agree once they agree on basis blades, where both reduce by the anticommutator (1) to the same determinant of metric factors). Applied to two pseudoscalars it gives $\lambda_T \lambda_{T'} = -\det(p_a p_b)_{a \in T, b \in T'}$, so the result is at most linear in the λ 's jointly, of the form $P + \sum_T \lambda_T Q^T$ claimed in the theorem. Since all λ_T flip sign together under spatial inversion while every inner product is unchanged, the single common orientation bit is the unique non-polynomial output (see Sec. 3 main text). $\square$

Action of $\text{Spin}^+(1, 3)$ on $\text{Cl}(1, 3)$. A rotor $R = \exp(B/2)$ with bivector generator $B \in \langle\text{Cl}(1, 3)\rangle_2$ lies in the even subalgebra $\text{Cl}^+(1, 3)$. Since $\tilde{B} = -B$ on any bivector, reversal of the exponential gives

$$\tilde{R} = \exp(\tilde{B}/2) = \exp(-B/2) = R^{-1},$$

and hence $R\tilde{R} = \tilde{R}R = 1$. The rotor sandwich

$$\text{Ad}_R : X \mapsto RX\tilde{R} \quad (\text{B.5})$$

is therefore an *inner algebra automorphism* of $\text{Cl}(1, 3)$. It is $\mathbb{R}$ -linear in X , and inserting the identity $\tilde{R}R = 1$ between adjacent factors yields multiplicativity under the geometric product:

$$\begin{aligned} \text{Ad}_R(XY) &= RXY\tilde{R} = RX(\tilde{R}R)Y\tilde{R} \\ &= (RX\tilde{R})(RY\tilde{R}) = \text{Ad}_R(X)\text{Ad}_R(Y). \end{aligned} \quad (\text{B.6})$$

On the grade-one generators of $\text{Cl}(1, 3)$ one has the standard covering identity

$$R\gamma_\mu\tilde{R} = \Lambda(R)^\nu{}_\mu\gamma_\nu \in \langle\text{Cl}(1, 3)\rangle_1, \quad (\text{B.7})$$

where $\Lambda(R) \in \text{SO}^+(1, 3)$ is the proper orthochronous Lorentz transformation associated with R through the two-to-one covering homomorphism $\text{Spin}^+(1, 3) \rightarrow \text{SO}^+(1, 3)$ [1]. Combining eqs. (B.6) and (B.7), on a generic r -blade $P_r = p_1 \wedge \cdots \wedge p_r$ built from grade-one elements $p_a = p_a^\mu \gamma_\mu$ we obtain

$$\text{Ad}_R(p_1 \wedge \cdots \wedge p_r) = (\Lambda p_1) \wedge \cdots \wedge (\Lambda p_r) \in \langle\text{Cl}(1, 3)\rangle_r : \quad (\text{B.8})$$

Ad_R acts on every grade-one factor as the Lorentz transformation Λ , and because it is an algebra automorphism it commutes with the antisymmetrisation built into the wedge product $p_1 \wedge \cdots \wedge p_r = \frac{1}{r!} \sum_\sigma \text{sgn}(\sigma) p_{\sigma(1)} \cdots p_{\sigma(r)}$, so the right-hand side of eq. (B.8) is again a grade- r blade. By $\mathbb{R}$ -linearity, Ad_R preserves every grade subspace $\langle\text{Cl}(1, 3)\rangle_k$ of $\text{Cl}(1, 3)$ (since each is the linear span of its r -blades).

On grade zero the action is trivial; on grade one it is the fundamental representation (B.7); on grade two it is the adjoint representation acting on the six-dimensional space of infinitesimal Lorentz transformations; on grades three and four it is determined via the Hodge isomorphisms $\langle\text{Cl}(1, 3)\rangle_3 \xrightarrow{I^{-1}} \langle\text{Cl}(1, 3)\rangle_1$ and $\langle\text{Cl}(1, 3)\rangle_4 \xrightarrow{I^{-1}} \langle\text{Cl}(1, 3)\rangle_0$, which intertwine the $\text{Spin}^+(1, 3)$ action because the pseudoscalar I commutes with every even-grade element of $\text{Cl}(1, 3)$ — in particular with any rotor R — so that $\text{Ad}_R(XI^{-1}) = \text{Ad}_R(X)I^{-1}$. The action on the event multivector $\mathcal{E}$ touches only the $\text{Cl}(1, 3)$ factor of eq. (4), leaving the object-type factor V_{flav} invariant.

C Tables

This appendix collects the two reference tables that back the content claims of the main text: the full per-grade dictionary of 32 classical observables (App. C.1) and the symmetry inventory split into four sub-tables (App. C.2).

C.1 Full per-grade dictionary of classical observables

Tab. 5 below catalogues 32 classical observables of hadron-collider events organised by the grade of the underlying multivector channel and by the algebraic operation that extracts them. The “in $\langle \mathcal{E} \rangle_0$?” column tracks which observables collapse, via Theorem 1, onto the inner-product (grade-zero) sector [4, 5].

Before turning to the dictionary itself, Tab. 4 records how the number of independent kinematic slots per grade of $\mathcal{E}$ scales with the final-state multiplicity N , taken as the number of final fermions after the decays $t \rightarrow bW$, $W \rightarrow f\bar{f}'$ (with neutrinos replaced by a single E_T^{miss} pseudo-particle, so N counts the fermion-level objects fed to the network rather than the reco-level jets). The counts are purely combinatorial: starting from the N slots of eq. (4), grade 1 is the set of N four-momenta, grade $k \in \{2, 3, 4\}$ collects the $\binom{N}{k}$ wedge products $p_{i_1} \wedge \cdots \wedge p_{i_k}$, and grade 0 contains the N on-shell masses m_i^2 together with the $\binom{N}{2}$ pairwise inner products $p_i \cdot p_j$.

N	grade 0 $\binom{N}{2}+N$	grade 1 $\binom{N}{1}$	grade 2 $\binom{N}{2}$	grade 3 $\binom{N}{3}$	grade 4 $\binom{N}{4}$	Total slots
<i>components / slot</i> = $\binom{4}{k}$:	1	4	6	4	1	
4 (minimal example)	10	4	6	4	1	25
6 (e.g. $t\bar{t}$ semileptonic)	21	6	15	20	15	77
12 (e.g. $t\bar{t}t\bar{t}$ all-hadronic)	78	12	66	220	495	871

Table 4: Number of independent kinematic *slots* per grade of $\mathcal{E}$ as a function of the final-state multiplicity N (fermion-level, $\nu \rightarrow E_T^{\text{miss}}$). One *slot* stands for *one* $\text{Cl}(1, 3)$ *multivector* of the indicated grade, attached to a specific subset of the N tokens (its V_{flav} label, $|f_{i_1}\rangle \otimes \cdots \otimes |f_{i_k}\rangle$, is inherited from those tokens and is not counted separately). Slot counts are purely combinatorial: $\#(\text{grade } 0) = N + \binom{N}{2}$ (on-shell masses plus pairwise inner products), $\#(\text{grade } 1) = N$ (four-momenta), $\#(\text{grade } k) = \binom{N}{k}$ for $k \in \{2, 3, 4\}$ (wedge products). Each slot carries $\binom{4}{k}$ real components, the dimension of the grade- k sector of $\text{Cl}(1, 3)$ (the second header row); the raw component count on grade k is therefore $(\text{slots}) \times \binom{4}{k}$, e.g. for $N = 6$: grade 1 = $6 \times 4 = 24$ numbers, grade 2 = $15 \times 6 = 90$, grade 3 = $20 \times 4 = 80$, total 230.

Two observations qualify these counts. *First*, $\text{Cl}(1, 3)$ has only finite per-grade dimension, $\dim\langle \text{Cl}(1, 3) \rangle_k = \binom{4}{k} \in \{1, 4, 6, 4, 1\}$ for $k = 0, \dots, 4$. For $N > 4$ the formal $\binom{N}{k}$ wedge products are therefore linearly dependent: they all live inside a $\binom{4}{k}$ -dimensional subspace, so the genuine geometric content (e.g. a single oriented 4-volume at grade 4) saturates already at $N = 4$. The extra entries label *which* k -tuple was chosen, not new geometry, and are exactly the λ_{ijkl} pseudoscalars whose redundancy underwrites Theorem 1. *Second*, a practical network does not consume the full combinatorial set: the attention layers select a physics-motivated subset (e.g. b -jets paired with their candidate W -decay products) or a top- K truncation by norm, since the unweighted full set scales as $O(N^4)$ while the resonance topology of tWb already fits in $O(N)$ tokens (Sec. 7).

Table 5: Full per-grade dictionary of 32 classical observables of hadron-collider events, organised by the grade of the underlying multivector channel of $\mathcal{E}$ and by the algebraic operation that extracts them. Citations mark entries with a specific originating work; the remainder are standard. Column “in $\langle \mathcal{E} \rangle_0$?” marks which observables collapse, via Theorem 1, onto the grade-zero scalar sector: “Y” = invariant reduction; “ H_{LHC} ” = reduction after the beam-axis tensor contraction; “N” = genuinely new (not in the grade-zero sector); “n.a.” = the object is not a blade in $\text{Cl}(1, 3)$ and lives in the orthogonal complement of $\bigwedge^\bullet(\mathbb{R}^{1,3})$.

#	Observable	Grade $\times$ operation	$\langle \mathcal{E} \rangle_0$?	Physical meaning
1	$\hat{s} = (\sum_i p_i)^2$	$0 \times \langle (\sum p)^2 \rangle_0$	Y	Partonic CM energy squared.
2	$\hat{t} = (p_a - p_c)^2$	$0 \times \langle \cdot \rangle_0$	Y	Mandelstam t -channel exchange.
3	$\hat{u} = (p_a - p_d)^2$	$0 \times \langle \cdot \rangle_0$	Y	Mandelstam u -channel exchange.
4	$m_S^2 = (\sum_{i \in S} p_i)^2$	$0 \times \langle \cdot \rangle_0$	Y	Invariant mass of any subset S ; W, Z, H , top and $t\bar{t}$ peaks.
5	$m_i^2 = p_i \cdot p_i$	$0 \times \langle p_i p_i \rangle_0$	Y	On-shell mass of object i .
6	$\cos \theta^*$ spin projections (helicity and optimal bases) [30–32]	0 via rotor sandwich + dot	Y	Decay-angle spin polarimeters in the top rest frame.
7	$\cos \theta_{\text{CS}}$ (Collins–Soper)	0 via rotor sandwich + dot	Y	Lepton-pair angle; A_{FB} sensitivity.
8	Dalitz-plot variables (m_{ij}^2, m_{jk}^2) [46]	$0 \times \langle (p_i + p_j)^2 \rangle_0$ on $\det G_3 = 0$	Y	3-body decay phase space; Cayley–Menger boundary.
9	$p_T^f, \eta^f, \phi^f, y^f, M_T^{f,a}$	1 tensored with $\hat{n}_z$	H_{LHC}	Lab-frame transverse momentum, pseudorapidity, azimuth, rapidity and transverse mass of object f ; $\text{SO}(2)_\phi \times \text{SO}^+(1, 1)_z$ basis.
10	$\Delta R_{ij} = \sqrt{\Delta \eta^2 + \Delta \phi^2}$	1 via beam-tensored differences	H_{LHC}	Catchment-cone distance.
11	$\Delta \phi_{ij}$	1 via B^{12} projection	H_{LHC}	Azimuthal opening angle.
12	Δy_{ij}	1 via B^{0z} projection	$\text{SO}^+(1, 1)_z$ -only	Rapidity difference; longitudinal-boost invariant.
13	$H_T = \sum_{\text{jets}} p_T$	1, sum of transverse norms	H_{LHC}	Hadronic activity scalar sum.
14	$S_T = H_T + \sum p_T^\ell + E_T^{\text{miss}}$	1, sum of transverse norms	H_{LHC}	Total transverse energy.
15	E_T^{miss} pseudo-particle	1 with masked p_z component	H_{LHC} -only	Sum of invisible transverse momenta.
16	$B_W = p_\ell \wedge p_\nu$ [9, 10]	2 via outer product	raw: N; $ B_W ^2$: Y	Oriented decay-plane bivector of W .
17	$B_t = (p_\ell + p_\nu) \wedge p_b$ [10]	2 via outer product	raw: N; $ B_t ^2$: Y	Oriented decay-plane bivector of t .
18	C_{ij} (Bernreuther basis k, r, n) [34, 35]	2 raw / 0 via $\langle B_a \widetilde{B}_b \rangle_0$	inv. proj.: Y; raw: N	$t\bar{t}$ spin correlation.
19	S^{ij} (sphericity tensor)	symmetric rank-2; not in $\bigwedge^2$	n.a.	Eigenspectrum of momentum tensor; null-row.
20	Spinor-helicity $\langle ij \rangle, [ij]$ [47–49]	$\text{Spin}^+(1, 3) \simeq \text{SL}(2, \mathbb{C})$ inside the even subalgebra $\text{Cl}^+(1, 3) \simeq M_2(\mathbb{C})$; not a blade in $\bigwedge^\bullet(\mathbb{R}^{1,3})$	n.a.	Massless QCD amplitude pair products; spinor lift of $\text{Spin}^+(1, 3)$.
21	S, A, P (sphericity, aplanarity, planarity)	0 via eigenvalues of S^{ij}	$\text{SO}(3)_{\text{lab}}$ only (not boost-inv.)	Event-shape scalars.

Continued on next page.

Tab. 5 continued from previous page.

#	Observable	Grade $\times$ operation	$\langle \mathcal{E} \rangle_0?$	Physical meaning
22	Thrust T	0 via $\max_{\hat{n}} \sum p_i \cdot \hat{n} $	$SO(3)_{\text{lab}}$ only (not boost-inv.)	Maximal alignment scalar.
23	Fox–Wolfram moments H_ℓ	0 via $\sum p_i p_j P_\ell(\cos \theta_{ij})$	$SO(3)_{\text{lab}}$ only (not boost-inv.)	Spherical-harmonic moments.
24	n -subjettiness τ_N [36]	0 via jet-axis projections	$SO(3)_{\text{lab}}$ only (not boost-inv.)	Jet-substructure axis-counting.
25	Soft-drop mass m_{SD} [37]	0 via (y, ϕ) jet-grooming	H_{LHC} -only	Groomed jet mass.
26	$\vec{p}_a \cdot (\vec{p}_b \times \vec{p}_c)$ [18]	3 via T^{123} component of $p_a \wedge p_b \wedge p_c$	sign: N; $ T ^2$: Y	P-odd triple product (motion-reversal-T-odd in the sense of ref. [18]; covariant-T-even, cf. Tab. 7); CP-violation probe.
27	$\det G_3 = \det[p_a \cdot p_b]_{a,b \in \{i,j,k\}}$	0 via $\langle T \tilde{T} \rangle_0$	Y	Squared 3-volume in Minkowski.
28	$\det G_4 = \det[p_a \cdot p_b]_{a,b \in \{i,j,k,l\}}$	0 via $\langle P \tilde{P} \rangle_0$	Y	Squared 4-volume magnitude.
29	$\text{sgn}\langle p_1 p_2 p_3 p_4 \rangle_4$ (this work) [18, 35]	4 via $\text{sgn}(\lambda)$, $\lambda I = p_1 \wedge \dots \wedge p_4$	N (genuinely new)	Sign of oriented 4-volume; CP-odd.
30	Bernreuther $T_5, B_3, \mathcal{A}_{CP}$ [35]	4 via sgn grade-4 / 3 raw	mostly via $\text{sgn}\langle \cdot \rangle_4$: N	CP-odd / T-odd correlations in $t\bar{t}$ decays.
31	Forward–backward asymmetry A_{FB}	1 via $\text{sgn}(p_z \cdot \hat{n}_{\text{dir}})$	H_{LHC}	$\hat{z}$ -projection sign asymmetry.
32	Charge-asymmetry $Q_\ell \eta_{\bar{j}}$ [50]	$V_{\text{flav}} \otimes \langle \text{Cl}(1, 3) \rangle_1$: Q_ℓ from V_{flav} , $\eta_{\bar{j}}$ from grade-1	H_{LHC}	Lepton-charge $\times$ forward-jet pseudorapidity; single-top tag.

^a For neural-network inputs we recommend the embedding $(\cos \phi^f, \sin \phi^f)$ in place of the raw angle.

Per-grade allocation summary. The 32 rows of Tab. 5 populate the grade decomposition of $\mathcal{E}$ as follows: rows 1–8, 21–25, 27–28 sit in the grade-zero (scalar) sector; rows 9–15 and 31 sit in the grade-one (four-vector) sector; row 32 sits in the V_{flav} -tensored grade-one (charge-weighted) sector; rows 16–18 sit in the grade-two (bivector) sector; row 26 in the grade-three sector; rows 29–30 in the grade-four (pseudoscalar) sector. Rows 19 and 20 are the sphericity tensor S^{ij} and the spinor-helicity pair $\langle ij \rangle, [ij]$; both objects are *not* blades in $\text{Cl}(1, 3)$ and sit in the orthogonal complement of $\bigwedge^\bullet(\mathbb{R}^{1,3})$. They are listed as null rows so that the two objects a reader may expect to find are accounted for: the symmetric rank-two momentum tensor, discussed in the event-shape caveat of Sec. 3, and the spinor-helicity brackets, which are bilinears in the even subalgebra $\text{Cl}^+(1, 3) \simeq M_2(\mathbb{C})$ rather than blades in $\bigwedge^\bullet(\mathbb{R}^{1,3})$.

C.2 Symmetry inventory

This appendix splits the 30 symmetries of $\mathcal{E}$ inventoried in Sec. 5 into four functional groups: continuous spacetime symmetries (Tab. 6), discrete spacetime symmetries (Tab. 7), internal and permutational symmetries (Tab. 8), and approximate, ML-architectural and CP-channel symmetries (Tab. 9).

Table 6: Continuous spacetime symmetries.

#	Name	Group	Action on $\text{Cl}(1, 3)$	On V_{flav}	Enforcement
A.1	Lorentz proper orthochr.	$\text{Spin}^+(1, 3) \simeq \text{SL}(2, \mathbb{C})$	$\mapsto R X \tilde{R}$ (rotor sandwich)	trivial	strict on parton; approx. on reco
A.2	Translation $T(4)$	$\mathbb{R}^4$ abelian	trivial on 4-momenta (Noether-conserved); $x_i \mapsto x_i + a$ on positions in PGA $\text{Cl}(1, 3, 1)$ (out of scope of $\mathcal{E}$)	trivial	out (mention only)
A.3	Conformal	$\text{SO}(2, 4)$, 15-D	$\text{Cl}(2, 4)$ extension	trivial	out (mention only)
A.4	Dilatation	$\text{SO}(1, 1)_D$	grade-preserving scale	trivial	soft (layer-norm)
A.5	Beam-axis residual H_{LHC}	$\text{SO}^+(1, 1)_\times \text{SO}(2)_\phi$	two rotor sandwiches	trivial	strict (incl. E_T^{miss})
A.6	2+2 algebraic split	$\text{Cl}(1, 1)_\times \hat{\otimes} \text{Cl}(0, 2)_\perp$	basis tensor decomposition	trivial	strict (basis choice)

C.2.1 Continuous spacetime symmetries

C.2.2 Discrete spacetime symmetries

Discrete spacetime symmetries (P , T , C , CP , CPT) act on $\mathcal{E}$ as outer automorphisms of $\text{Cl}(1, 3) \otimes V_{\text{flav}}$; Tab. 7 collects their algebraic realisation.

Two conventions for T . Two definitions of the Lorentz-improper time-reversal element are in common use, and the present paper consciously employs both labels in different contexts. The *Wigner motion-reversal* convention $\Lambda_T^W = \text{diag}(+1, -1, -1, -1)$ — adopted by Atwood, Bernreuther et al. to define “T-odd triple products” [18, 34, 35] — acts on 4-momenta as $(E, \vec{p}) \rightarrow (E, -\vec{p})$, and as a matrix on 4-vectors coincides with parity P ; the two are distinguished only by the antiunitarity of T . The *covariant T* convention $\Lambda_T^{\text{cov}} = \text{diag}(-1, +1, +1, +1)$, adopted in Sec. 5 of this paper and explicitly invoked in the Grade 4 paragraph of Sec. 3 (around eq. (8)), instead sends $(E, \vec{p}) \rightarrow (-E, +\vec{p})$: it flips the time-component and leaves the 3-momentum unchanged. As 4×4 matrices on 4-vectors the two are related by an overall sign, $\Lambda_T^{\text{cov}} = -\Lambda_T^W$, and both have $\det = -1$, so they agree on every Lorentz-improper one-bit channel. In particular, the grade-four pseudoscalar $\lambda = \epsilon_{\mu\nu\rho\sigma} p_i^\mu p_j^\nu p_k^\rho p_l^\sigma$ (eq. (8)) is T-odd in either convention, so the CP-odd channel $\text{sgn } \lambda$ implemented by the architecture (row G.1 of Tab. 9) is convention-independent. The two conventions *disagree*, however, on the T-parity of grade-three spatial-only observables — the triple product $\vec{p}_a \cdot (\vec{p}_b \times \vec{p}_c)$ is motion-reversal-T-odd (its three spatial momenta each flip sign under Λ_T^W) but covariant-T-even (its spatial components are preserved under Λ_T^{cov}); it is P-odd in both conventions. Tab. 7 below realises T in the covariant convention, while Tab. 5 retains the standard motion-reversal “T-odd” label of refs. [18, 35] on the triple-product row of Tab. 5 (with an inline disambiguation). Other approaches in the literature, including $\gamma_T = \gamma_0$ (Wigner-T as antiunitary parity, equivalent on 4-momentum to row B.1) and $\gamma_T = i\gamma_1\gamma_3$ (Dirac-spinor Wigner-T, which leaves $\text{Cl}(1, 3)$ real only after i is absorbed into the antiunitary factor), are equivalent up to relabelings and antiunitary signs; the covariant choice is taken here because it makes Λ_T^{cov} act non-trivially on the time-axis generator γ_0 and trivially on γ_i , so that the T-channel of the architecture decouples from the parity channel B.1 at the level of the Clifford action.

C.2.3 Internal and permutational symmetries

The internal gauge symmetries acting on V_{flav} and the combinatorial S_n permutations of the per-object tokens are collected in Tab. 8.

Table 7: Discrete spacetime symmetries (T realised in the covariant convention, see the discussion of two conventions for T above). The time-reversal element is $\gamma_T \equiv \gamma_1 \gamma_2 \gamma_3$, the spatial pseudoscalar of $\text{Cl}(1, 3)$ (real-valued, with $\gamma_T^2 = +1$ in the $(+, -, -, -)$ signature, so $\gamma_T^{-1} = \gamma_T$); the antiunitary action of row B.2 includes complex conjugation (“c.c.”); C_V denotes the action of charge conjugation on the object-type factor V_{flav} , exchanging particle/antiparticle labels.

#	Name	Group	Action on $\text{Cl}(1, 3)$	On V_{flav}	Enforcement
B.1	Parity P	$\mathbb{Z}_2$	$X \mapsto \gamma_0 X \gamma_0$	trivial	not enforced (preserves CP-odd channel)
B.2	Time reversal T	$\mathbb{Z}_2$ antiunitary	$X \mapsto \gamma_T X \gamma_T^{-1} + \text{c.c.}$ ($\gamma_T^2 = +1$, so $\gamma_T^{-1} = \gamma_T$; covariant-T convention, $\Lambda_T^{\text{cov}} = \text{diag}(-1, +1, +1, +1)$)	trivial	not enforced (broken via CPT $\oplus$ CP)
B.3	Charge conjugation C	$\mathbb{Z}_2$	trivial	$f \rightarrow C_V f$, $f^+ \leftrightarrow f^-$	not enforced (charge-bit embedding)
B.4	CP	$\mathbb{Z}_2$	$X \mapsto \gamma_0 X \gamma_0$	$f \rightarrow C_V f$	not enforced (CP-asymmetry head reads the channel, Sec. 8)
B.5	CPT	identity (group)	$X \mapsto \hat{X} + \text{c.c.}$ (grade-involution $(-1)^r$ on grade- r from $P \cdot T$; trivial on physical bilinears by the CPT theorem)	$f \rightarrow C_V f$ (inherited from C)	automatic (CPT theorem; not implemented)

Table 8: Internal and permutational symmetries.

#	Name	Group	$\text{Cl}(1, 3)$	Action on V_{flav}	Enforcement
C.1	$U(1)_{\text{em}}$	$U(1)$	trivial	charge-bit on V_{flav}	embedding (active)
C.2	EW $SU(2)_L \times U(1)_Y$	4-D	trivial	not in feature space	out (handled by MC)
C.3	Colour $SU(3)_C$	8-D	trivial	not in feature space	out (confined; MC)
C.4	CKM unitarity	$U(3)/\text{phases}$ (4 params)	trivial	not in feature space	out (handled by MC)
C.5	PMNS unitarity	$U(3)/\text{phases}$ (4 or 6 params)	trivial	not in feature space	out (handled by MC)
D.1	S_n within flavour type	$\prod_a S_{n_a}$	trivial	tensor-symmetric on V_{flav}	strict (set-attention)
D.2	Bose/Fermi statistics	$\mathbb{Z}_2$ amplitude	trivial	implicit in MC	covered by D.1
D.3	LFU S_3	S_3 on $e/\mu/\tau$	trivial	weight-tying option	soft (document break by masses)
D.4	Generation (quarks)	S_3 S_3	trivial	strongly broken	not enforced (Yukawa)
D.5	Isospin $SU(2)_I$	3-D	trivial	confined hadron multiplets	out (broken by EW + masses)

C.2.4 Approximate, ML-architectural and CP-channel symmetries

Tab. 9 lists the softer symmetries that shape the architecture without entering as hard equivariance constraints (dilatation, chiral and custodial symmetries, the absence of positional en-

coding, and the CP-odd pseudoscalar-sign channel).

Table 9: Approximate, ML-architectural and CP-channel symmetries.

#	Name	Group	Cl(1, 3)	Action on V_{flav}	Enforcement
E.1	Chiral $SU(N)_L \times SU(N)_R$	$2(N^2-1)$	trivial	trivial	out (confined; $S\chi$ SB)
E.2	Heavy-quark sym- metry	$SU(2)_s \times U(N_h)_f$ (N_h heavy flavours, e.g. b, c)	trivial	trivial	out (top decays first)
E.3	Custodial $SU(2)_V$	3-D	trivial	trivial	out (Higgs- sector specific)
F.1	Architectural set- perm.	S_n	trivial	tensor-symmetric (set- attention)	strict (by con- struction)
F.2	Token positional shift	$\mathbb{Z}_n$	trivial	none (no positional en- coding)	conditional (absent-by- design)
F.3	Feature-scale	$\mathbb{R}_+$	trivial	layer-norm	soft (architec- tural)
G.1	Pseudoscalar sign $\mathbb{Z}_2$	$\mathbb{Z}_2$	$\text{sgn}(p_1 p_2 p_3 p_4)_4 \rightarrow -\text{sgn}$	trivial	conditional (CP- asymmetry head input, Sec. 8)
G.2	$\mathcal{A}_{CP}$ in decay rates	$\mathbb{Z}_2$	trivial	charge-bit symmetrisation	anti-embedding (implicit)
G.3	EDMs	$\mathbb{Z}_2$	trivial	trivial	out (low- energy, not LHC)

D Token layouts of the two input branches

A token (defined at first use in Sec. 1) is a stack of *channels*, each channel a full 16-component multivector of $\text{Cl}(1, 3)$. The $1+4+6+4+1$ decomposition of those 16 components is the basis of the algebra — one scalar cell, the four γ_μ , the six coordinate planes $\gamma_{\mu\nu}$, the four $\gamma_{\mu\nu\rho}$, and the pseudoscalar γ_{0123} — i.e. storage for the five grades, not content: a single four-momentum populates only the four grade-one cells, and every nonzero higher-grade object requires several particles, $p_i \wedge p_i = 0$. The equivariant linear layers mix channels with one weight per grade (five per channel pair), which is the entire 2080-parameter difference between the two branches of Tab. 3 (13 extra input channels $\times$ 32 outputs $\times$ 5).

One bookkeeping remark, since the layouts below look at first sight unlike eq. (4). A slot $p_i \otimes |f_i\rangle$ is a *simple* tensor, and the tables store its two factors separately — the momentum in the grade-one cells of one channel, the label as a one-hot across grade-zero cells — rather than as the $4 \times \dim V_{\text{flav}}$ dense array of coefficients. For a simple tensor with $|f_i\rangle$ a basis vector the two are in bijection, so nothing is lost; the dense form would only be needed for superpositions of type labels, which no reconstruction produces.

Four-momentum branch (grade $0 \oplus 1$; 6–8 tokens, 9 channels). A particle token fills 12 of its 9×16 cells:

channel	grade cells	content	meaning
0	1 (four cells)	$(E, p_x, p_y, p_z)_i$	measured momentum
1–6	0 (one cell each)	type one-hot	object identity
7	0	electric charge	
8	0	b -tag	

All grade-2,3,4 cells enter as zeros; the network populates them itself — attention mixes momenta across tokens with Lorentz-invariant weights, and the geometric-product layer then forms $p_i \wedge p_j$ (grade 2), triple products (grade 3) and beyond within a token, so that L blocks generate products of depth $\sim 2^L$: the closure of Sec. 2.2, realised by the network dynamics. The reference rows of Tab. 3 append one or two further tokens of the same shape, each carrying one fixed element of the algebra in its covariant channel: two vector tokens for γ_0 and γ_3 , or a single token with the grade-two cell γ_{03} set. Constants need no trainable embedding, which is why those rows share one parameter count.

Pairing branch (8 tokens, 22 channels). The six particle tokens are as above inside a wider layout (channel 1 and the ten pairing-scalar channels zero for particles; the type one-hot spans eight slots, the six particle species plus one sentinel slot per pairing token). The two event-level pairing tokens, one per candidate b -assignment $a \in \{1, 2\}$, arrive with grades zero to three pre-filled by the topology candidates:

channel	grade cells	content	meaning
0	1 (four)	$\star T_{\text{had}}^{(a)}$	dual of hadronic-top volume
0	2 (six)	$T_{\text{had}}^{(a)} \cap T_{\text{lep}}^{(a)}$	meet of the two decay volumes
0	3 (four)	$T_{\text{had}}^{(a)} = p_{j_1} \wedge p_{j_2} \wedge p_{b_a}$	hadronic-top candidate volume
1	$1 \oplus 3$	$\star T_{\text{lep}}^{(a)}, T_{\text{lep}}^{(a)}$	leptonic side
12–21	0	pulls, $\sigma_{Wb}^\pm, m_W$, coplanarity, $\text{sgn } \lambda$	topology invariants

In both branches the event is one structure — slots $\times$ channels $\times$ 16 grade cells — and the branches differ only in which cells arrive populated. Materialising *all* derived components explicitly ($\binom{N}{2}$ bivectors, $\binom{N}{3}$ trivectors, $\binom{N}{4}$ pseudoscalar coefficients, each as its own token) is the upper rung of the ladder of Sec. 2.2, implemented neither here nor elsewhere; the pairing tokens materialise only the components tied to the resonance-topology hypotheses.

References

- [1] C. Doran and A. Lasenby, *Geometric Algebra for Physicists*, Cambridge University Press, doi:[10.1017/CBO9780511807497](https://doi.org/10.1017/CBO9780511807497) (2003).
- [2] D. Hestenes and G. Sobczyk, *Clifford Algebra to Geometric Calculus: A Unified Language for Mathematics and Physics*, Reidel, Dordrecht, doi:[10.1007/978-94-009-6292-7](https://doi.org/10.1007/978-94-009-6292-7) (1984).
- [3] P. Lounesto, *Clifford Algebras and Spinors*, vol. 286 of *London Mathematical Society Lecture Note Series*, Cambridge University Press, 2 edn., ISBN 9780521005517, doi:[10.1017/CBO9780511526022](https://doi.org/10.1017/CBO9780511526022) (2001).
- [4] E. E. Boos, V. E. Bunichev, L. V. Dudko and A. A. Markina, *Method of “optimum observables” and implementation of neural networks in physics investigations*, Phys. Atom. Nucl. **71**(2), 388 (2008), doi:[10.1134/s1063778808020191](https://doi.org/10.1134/s1063778808020191).

- [5] L. Dudko, G. Vorotnikov, P. Volkov, D. Ovchinnikov, M. Perfilov, A. Shporin and A. Chernodod, *General recipe to form input space for deep learning analysis of HEP scattering processes*, Int. J. Mod. Phys. A **35**(21), 2050119 (2020), doi:[10.1142/S0217751X20501195](https://doi.org/10.1142/S0217751X20501195), [2002.09350](https://arxiv.org/abs/2002.09350).
- [6] A. Bogatskiy, B. Anderson, J. T. Offermann, M. Roussi, D. W. Miller and R. Kondor, *Lorentz Group Equivariant Neural Network for Particle Physics* (2020), [2006.04780](https://arxiv.org/abs/2006.04780).
- [7] A. Bogatskiy, T. Hoffman, D. W. Miller and J. T. Offermann, *PELICAN: Permutation Equivariant and Lorentz Invariant or Covariant Aggregator Network for Particle Physics* (2022), [2211.00454](https://arxiv.org/abs/2211.00454).
- [8] J. Brehmer, P. de Haan, S. Behrends and T. Cohen, *Geometric Algebra Transformer*, In *Advances in Neural Information Processing Systems*, vol. 37 (2023), [2305.18415](https://arxiv.org/abs/2305.18415).
- [9] J. Spinner, V. Bresó, P. de Haan, T. Plehn, J. Thaler and J. Brehmer, *Lorentz-Equivariant Geometric Algebra Transformers for High-Energy Physics*, In *38th conference on Neural Information Processing Systems* (2024), [2405.14806](https://arxiv.org/abs/2405.14806).
- [10] J. Brehmer, V. Bresó, P. de Haan, T. Plehn, H. Qu, J. Spinner and J. Thaler, *A Lorentz-equivariant transformer for all of the LHC*, SciPost Phys. **19**(4), 108 (2025), doi:[10.21468/SciPostPhys.19.4.108](https://doi.org/10.21468/SciPostPhys.19.4.108), [2411.00446](https://arxiv.org/abs/2411.00446).
- [11] E. E. Boos, V. E. Bunichev, P. V. Volkov, L. V. Dudko and M. A. Perfilov, *Separation of Pair and Single Top Quark Production in tWb Associated Final State Using a Neural Network*, Moscow Univ. Phys. Bull. **78**(6), 707 (2023), doi:[10.3103/S0027134923060024](https://doi.org/10.3103/S0027134923060024).
- [12] E. E. Boos, V. E. Bunichev, L. V. Dudko and M. A. Perfilov, *Application of the Subsidiary Fields Method to the Modeling of $tW + t\bar{t}$ Processes with the Anomalous Wtb Interactions*, Phys. Atom. Nucl. **83**(6), 989 (2020), doi:[10.1134/S1063778820060095](https://doi.org/10.1134/S1063778820060095).
- [13] A. Baskakov, E. Boos, V. Bunichev, L. Dudko, M. Perfilov and P. Volkov, *Recommendations for the search of the Anomalous Wtb interactions in the tW -associated Single Top Quark Production*, EPJ Web Conf. **222**, 04010 (2019), doi:[10.1051/epjconf/201922204010](https://doi.org/10.1051/epjconf/201922204010).
- [14] P. de Haan, T. Cohen and J. Brehmer, *Euclidean, Projective, Conformal: Choosing a Geometric Algebra for Equivariant Transformers*, In *International Conference on Artificial Intelligence and Statistics* (2024), [2311.04744](https://arxiv.org/abs/2311.04744).
- [15] A. Cayley, *A theorem in the geometry of position*, Cambridge Math. J. **2**, 267 (1841).
- [16] K. Menger, *Untersuchungen über allgemeine Metrik*, Math. Ann. **100**, 75 (1928), doi:[10.1007/BF01448840](https://doi.org/10.1007/BF01448840).
- [17] K. Tate and M. Visser, *Realizability of the Lorentzian $(n,1)$ -Simplex*, JHEP **01**, 028 (2012), doi:[10.1007/JHEP01\(2012\)028](https://doi.org/10.1007/JHEP01(2012)028), [1110.5694](https://arxiv.org/abs/1110.5694).
- [18] D. Atwood, S. Bar-Shalom, G. Eilam and A. Soni, *CP violation in top physics*, Phys. Rept. **347**, 1 (2001), doi:[10.1016/S0370-1573\(00\)00112-5](https://doi.org/10.1016/S0370-1573(00)00112-5), [hep-ph/0006032](https://arxiv.org/abs/hep-ph/0006032).
- [19] V. Mikuni and B. Nachman, *Solving key challenges in collider physics with foundation models*, Phys. Rev. D **111**(5), L051504 (2025), doi:[10.1103/PhysRevD.111.L051504](https://doi.org/10.1103/PhysRevD.111.L051504), [2404.16091](https://arxiv.org/abs/2404.16091).
- [20] A. Bogatskiy, T. Hoffman, D. W. Miller, J. T. Offermann and X. Liu, *Explainable equivariant neural networks for particle physics: PELICAN*, JHEP **03**, 113 (2024), doi:[10.1007/JHEP03\(2024\)113](https://doi.org/10.1007/JHEP03(2024)113), [2307.16506](https://arxiv.org/abs/2307.16506).

- [21] S. Gong, Q. Meng, J. Zhang, H. Qu, C. Li, S. Qian, W. Du, Z.-M. Ma and T.-Y. Liu, *An efficient Lorentz equivariant graph neural network for jet tagging*, JHEP **07**, 030 (2022), doi:[10.1007/JHEP07\(2022\)030](https://doi.org/10.1007/JHEP07(2022)030), [2201.08187](https://arxiv.org/abs/2201.08187).
- [22] D. Ruhe, J. Brandstetter and P. Forré, *Clifford Group Equivariant Neural Networks*, In *Advances in Neural Information Processing Systems*, vol. 37 (2023), [2305.11141](https://arxiv.org/abs/2305.11141).
- [23] V. G. Satorras, E. Hoogeboom and M. Welling, *E(n) Equivariant Graph Neural Networks*, In M. Meila and T. Zhang, eds., *Proceedings of the 38th International Conference on Machine Learning*, vol. 139 of *Proceedings of Machine Learning Research*, pp. 9323–9332. PMLR (2021), [2102.09844](https://arxiv.org/abs/2102.09844).
- [24] P. T. Komiske, E. M. Metodiev and J. Thaler, *Energy Flow Networks: Deep Sets for Particle Jets*, JHEP **01**, 121 (2019), doi:[10.1007/JHEP01\(2019\)121](https://doi.org/10.1007/JHEP01(2019)121), [1810.05165](https://arxiv.org/abs/1810.05165).
- [25] H. Qu and L. Gouskos, *ParticleNet: Jet Tagging via Particle Clouds*, Phys. Rev. D **101**(5), 056019 (2020), doi:[10.1103/PhysRevD.101.056019](https://doi.org/10.1103/PhysRevD.101.056019), [1902.08570](https://arxiv.org/abs/1902.08570).
- [26] H. Qu, C. Li and S. Qian, *Particle Transformer for Jet Tagging* (2022), [2202.03772](https://arxiv.org/abs/2202.03772).
- [27] Y. Wu, K. Wang, C. Li, H. Qu and J. Zhu, *Jet tagging with more-interaction particle transformer**, Chin. Phys. C **49**(1), 013110 (2025), doi:[10.1088/1674-1137/ad7f3d](https://doi.org/10.1088/1674-1137/ad7f3d), [2407.08682](https://arxiv.org/abs/2407.08682).
- [28] M. J. Fenton, A. Shmakov, T.-W. Ho, S.-C. Hsu, D. Whiteson and P. Baldi, *Permutationless many-jet event reconstruction with symmetry preserving attention networks*, Phys. Rev. D **105**(11), 112008 (2022), doi:[10.1103/PhysRevD.105.112008](https://doi.org/10.1103/PhysRevD.105.112008), [2010.09206](https://arxiv.org/abs/2010.09206).
- [29] A. Shmakov, M. J. Fenton, T.-W. Ho, S.-C. Hsu, D. Whiteson and P. Baldi, *SPANet: Generalized permutationless set assignment for particle physics using symmetry preserving attention*, SciPost Phys. **12**(5), 178 (2022), doi:[10.21468/SciPostPhys.12.5.178](https://doi.org/10.21468/SciPostPhys.12.5.178), [2106.03898](https://arxiv.org/abs/2106.03898).
- [30] G. Mahlon and S. J. Parke, *Angular correlations in top quark pair production and decay at hadron colliders*, Phys. Rev. D **53**, 4886 (1996), doi:[10.1103/PhysRevD.53.4886](https://doi.org/10.1103/PhysRevD.53.4886), [hep-ph/9512264](https://arxiv.org/abs/hep-ph/9512264).
- [31] G. Mahlon and S. J. Parke, *Improved spin basis for angular correlation studies in single top quark production at the Tevatron*, Phys. Rev. D **55**, 7249 (1997), doi:[10.1103/PhysRevD.55.7249](https://doi.org/10.1103/PhysRevD.55.7249), [hep-ph/9611367](https://arxiv.org/abs/hep-ph/9611367).
- [32] G. Mahlon and S. J. Parke, *Single top quark production at the LHC: Understanding spin*, Phys. Lett. B **476**, 323 (2000), doi:[10.1016/S0370-2693\(00\)00149-0](https://doi.org/10.1016/S0370-2693(00)00149-0), [hep-ph/9912458](https://arxiv.org/abs/hep-ph/9912458).
- [33] H. Weyl, *The Classical Groups: Their Invariants and Representations*, Princeton University Press, Princeton, NJ (1939).
- [34] W. Bernreuther and Z.-G. Si, *Top quark spin correlations and polarization at the LHC: standard model predictions and effects of anomalous top chromo moments*, Phys. Lett. B **725**, 115 (2013), doi:[10.1016/j.physletb.2013.06.051](https://doi.org/10.1016/j.physletb.2013.06.051), [Erratum: Phys.Lett.B 744, 413–413 (2015)], [1305.2066](https://arxiv.org/abs/1305.2066).
- [35] W. Bernreuther, D. Heisler and Z.-G. Si, *A set of top quark spin correlation and polarization observables for the LHC: Standard Model predictions and new physics contributions*, JHEP **12**, 026 (2015), doi:[10.1007/JHEP12\(2015\)026](https://doi.org/10.1007/JHEP12(2015)026), [1508.05271](https://arxiv.org/abs/1508.05271).

- [36] J. Thaler and K. Van Tilburg, *Identifying Boosted Objects with N -subjettiness*, JHEP **03**, 015 (2011), doi:[10.1007/JHEP03\(2011\)015](https://doi.org/10.1007/JHEP03(2011)015), [1011.2268](https://arxiv.org/abs/1011.2268).
- [37] A. J. Larkoski, S. Marzani, G. Soyez and J. Thaler, *Soft Drop*, JHEP **05**, 146 (2014), doi:[10.1007/JHEP05\(2014\)146](https://doi.org/10.1007/JHEP05(2014)146), [1402.2657](https://arxiv.org/abs/1402.2657).
- [38] E. Boos and L. Dudko, *Triple top quark production in standard model*, Int. J. Mod. Phys. A **37**(05), 2250023 (2022), doi:[10.1142/S0217751X22500233](https://doi.org/10.1142/S0217751X22500233), [2107.07629](https://arxiv.org/abs/2107.07629).
- [39] A. Hayrapetyan et al., *Observation of a pseudoscalar excess at the top quark pair production threshold*, Rept. Prog. Phys. **88**(8), 087801 (2025), doi:[10.1088/1361-6633/adf7d3](https://doi.org/10.1088/1361-6633/adf7d3), [2503.22382](https://arxiv.org/abs/2503.22382).
- [40] G. Aad et al., *Observation of a cross-section enhancement near the $t\bar{t}$ production threshold in $\sqrt{s} = 13$ TeV pp collisions with the ATLAS detector* (2026), [2601.11780](https://arxiv.org/abs/2601.11780).
- [41] G. Aad et al., *Study of $t\bar{t}$ threshold effects in $e\mu$ differential distributions measured in $\sqrt{s} = 13$ TeV pp collisions with the ATLAS detector* (2026), [2605.02341](https://arxiv.org/abs/2605.02341).
- [42] E. Boos, V. Bunichev, M. Dubinin, L. Dudko, V. Ilyin, A. Kryukov, V. Edneral, V. Savrin, A. Semenov and A. Sherstnev, *CompHEP 4.4: Automatic computations from Lagrangians to events*, Nucl. Instrum. Meth. A **534**, 250 (2004), doi:[10.1016/j.nima.2004.07.096](https://doi.org/10.1016/j.nima.2004.07.096), [hep-ph/0403113](https://arxiv.org/abs/hep-ph/0403113).
- [43] *HiGEN: companion code repository for the present paper*, <https://github.com/lev-dudko/higen> (2026).
- [44] T. Golling, L. Heinrich, M. Kagan, S. Klein, M. Leigh, M. Osadchy and J. A. Raine, *Masked particle modeling on sets: towards self-supervised high energy physics foundation models*, Mach. Learn. Sci. Tech. **5**(3), 035074 (2024), doi:[10.1088/2632-2153/ad64a8](https://doi.org/10.1088/2632-2153/ad64a8), [2401.13537](https://arxiv.org/abs/2401.13537).
- [45] B. M. Dillon, G. Kasieczka, H. Olschlager, T. Plehn, P. Sorrenson and L. Vogel, *Symmetries, safety, and self-supervision*, SciPost Phys. **12**(6), 188 (2022), doi:[10.21468/SciPostPhys.12.6.188](https://doi.org/10.21468/SciPostPhys.12.6.188), [2108.04253](https://arxiv.org/abs/2108.04253).
- [46] R. H. Dalitz, *On the analysis of τ -meson data and the nature of the τ -meson*, Phil. Mag. Ser. 7 **44**, 1068 (1953), doi:[10.1080/14786441008520365](https://doi.org/10.1080/14786441008520365).
- [47] L. J. Dixon, *Calculating scattering amplitudes efficiently*, In *Theoretical Advanced Study Institute in Elementary Particle Physics (TASI 95): QCD and Beyond*, pp. 539–584 (1996), [hep-ph/9601359](https://arxiv.org/abs/hep-ph/9601359).
- [48] M. L. Mangano and S. J. Parke, *Multiparton amplitudes in gauge theories*, Phys. Rept. **200**, 301 (1991), doi:[10.1016/0370-1573\(91\)90091-Y](https://doi.org/10.1016/0370-1573(91)90091-Y).
- [49] D. Maitre and P. Mastrolia, *S@M, a Mathematica Implementation of the Spinor-Helicity Formalism*, Comput. Phys. Commun. **179**, 501 (2008), doi:[10.1016/j.cpc.2008.05.002](https://doi.org/10.1016/j.cpc.2008.05.002), [0710.5559](https://arxiv.org/abs/0710.5559).
- [50] J. A. Aguilar-Saavedra, *Single top quark production at LHC with anomalous Wtb couplings*, Nucl. Phys. B **804**, 160 (2008), doi:[10.1016/j.nuclphysb.2008.06.013](https://doi.org/10.1016/j.nuclphysb.2008.06.013), [0803.3810](https://arxiv.org/abs/0803.3810).